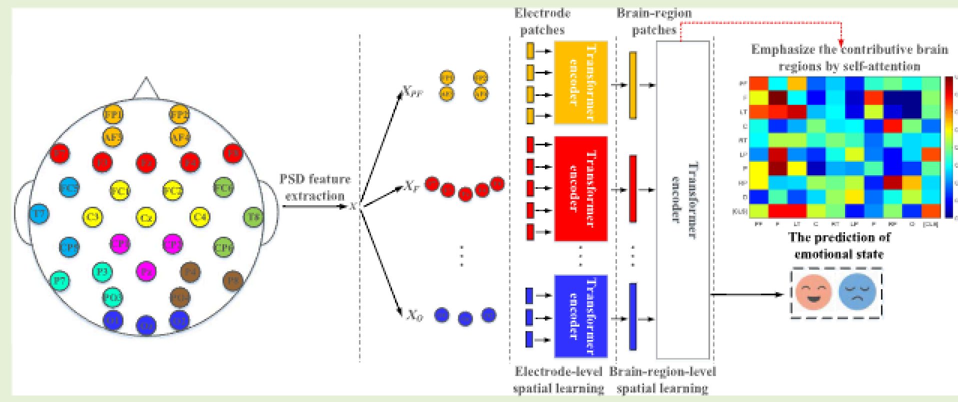
Transformers for EEG-Based Emotion Recognition: A Hierarchical Spatial Information Learning Model

Zhe Wang, Yongxiong Wang, Chuanfei Hu, Zhong Yin, and Yu Song, *Member, IEEE*

***Abstract*—The spatial information of Electroencephalography (EEG) is essential for emotion recognition model to learn discriminative feature. The convolutional networks and recurrent networks are the conventional choices to learn the complex spatial dependencies through a number of electrodes and brain regions. However, these models have difficulty in capturing long-range dependencies due to the operations of local feature learning. To enhance EEG spatial dependencies capturing and improve the accuracy of emotion recognition, we propose a transformer- based model to hierarchically learn the discriminative spatial information from electrode level to brain-region-level. In the electrode-level spatial learning, the transformer encoders are adopted to integrate information within different brain regions. Next, in view of the different roles of brain regions in the emotion recognition, the self-attention within the transformer could emphasize the contributive brain regions. Hence, in the brain-region-level spatial learning, a transformer encoder is utilized to capture the spatial dependencies among the brain regions. Finally, to validate the effectiveness of the proposed model, the subject-independent experiments are conducted on the DEAP and MAHNOB-HCI database. The experimental results demonstrate that the proposed model achieves outstanding performance in emotion recognition with arousal and valence level. Moreover, the visualization of self-attention indicates that the proposed model could emphasize the discriminative spatial information from pre-frontal lobe, frontal lobe, temporal lobe and parietal lobe.**



***Index Terms*— Emotion recognition, EEG, transformer, self-attention.**

I. Introduction

WITH the development of non-invasive sensor technology, the applications of physiological signals have attracted much attention [1]. Affective computing, which aims to build a harmonious human-computer interaction by endowing computers with the ability to recognize and comprehend human emotions [2], is an essential research direction among these applications. EEG [3]–[5], electromyography [6], and electro-cardiography [7] are the typical choices of physiological signals for emotion recognition. Among these modalities, EEG signals could record the activities of amygdala which is closely related to the emotions [8]. Hence, EEG-based emotion recognition has become a hot research topic in the fields of neuroscience and computer science.

In the machine learning based EEG emotion recognition methods, temporal and spectral features are commonly utilized to classify the emotions. Wang *et al.* [9] extract non-linear dynamic EEG features from time domain, such as approximate entropy and Hurst exponent. Kroupi *et al.* [10] extract Power Spectral Density (PSD) and Wasserstein distance from different frequency bands to identify emotions. Khare *et al.* [11] propose smoothed pseudo-Wigner–Ville distribution to generate the time-frequency representation which is learned by Convolutional Neural Network (CNN). These studies have made great progress to advance EEG-based emotion recognition, but many researchers do not take full advantage of the spatial dependencies among the electrodes or brain regions, denoted as the spatial information of EEG. The research of neuroscience indicates that the human emotion is related to the pre-frontal lobe [12] and parietal lobe [13]

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基于脑电图的情绪识别的 Transformer：一种分层空间信息学习模型

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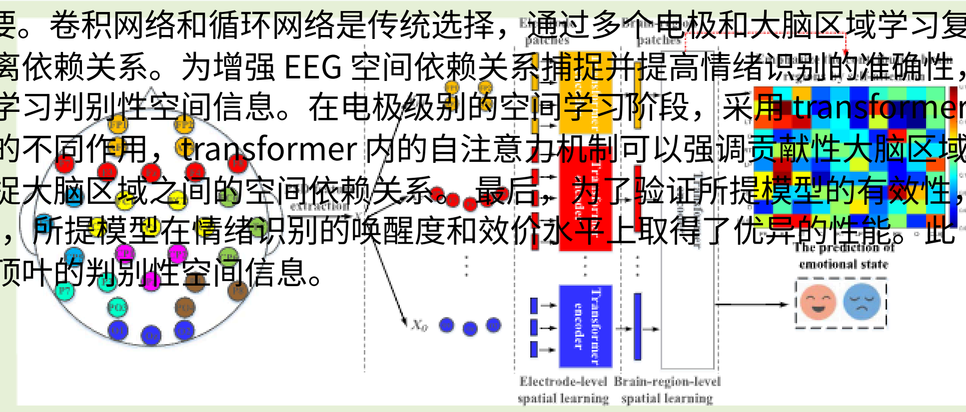
摘要—脑电图（EEG）的空间信息对于情绪识别模型学习判别性特征至关重要。卷积网络和循环网络是传统选择，通过多个电极和大脑区域学习复杂的空间依赖关系。然而，由于局部特征学习的操作，这些模型难以捕捉长距离依赖关系。为增强 EEG 空间依赖关系捕捉并提高情绪识别的准确性，我们提出了一种基于 transformer 的模型，从电极级别到大脑区域级别分层学习判别性空间信息。在电极级别的空间学习阶段，采用 transformer 编码器整合不同大脑区域内的信息。接下来，考虑到大脑区域在情绪识别中的不同作用，transformer 内的自注意力机制可以强调贡献性大脑区域。因此，在大脑区域级别的空间学习阶段，利用一个 transformer 编码器来捕捉大脑区域之间的空间依赖关系。最后，为了验证所提模型的有效性，在 DEAP 和 MAHNOBHCI 数据库上进行了与被试无关的实验。实验结果表明，所提模型在情绪识别的唤醒度和效价水平上取得了优异的性能。此外，自注意力机制的可视化表明，所提模型能够强调前额叶、额叶、颞叶和顶叶的判别性空间信息。



关键词—情绪识别，脑电图，Transformer，自注意力。

I. I

随着无创传感器技术的发展，生理信号的应用引起了广泛关注[1]。情感计算，旨在赋予计算机识别和理解人类情绪的能力，从而建立和谐的人机交互[2]，是这些应用中的重要研究方向。脑电图[3]–[5]、肌电图等图象[6]，以及心电图[7]是用于情绪识别的典型生理信号。在这些模态中，脑电图信号可以记录与情绪密切相关的杏仁核的活动[8]。因此，基于脑电图的情绪识别已成为神经科学和计算机科学领域的一个热门研究课题。

在基于机器学习的脑电图情绪识别方法中，通常利用时域和频域特征来分类情绪。王等人[9]从时域中提取非线性动态脑电图特征，如近似熵和赫斯特指数。Kroupi 等人[10]从不同频段提取功率谱密度（PSD）和 Wasserstein 距离来识别情绪。Khare 等人[11]提出平滑伪维格纳-维莱分布来生成时频表示，该表示由卷积神经网络（CNN）学习。这些研究在推进基于脑电图的情绪识别方面取得了巨大进展，但许多研究人员没有充分利用电极或脑区之间的空间依赖性，即脑电图的空间信息。神经科学的研究表明，人类情绪与前额叶[12]和顶叶[13]有关。

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where these brain regions could provide more contributive information than the others. What's more, Sangnark *et al.* [14] and Lakhan *et al.* [5] find the contributive brain regions or electrodes could achieve better performance than the others by analyzing the EEG response to the popular music and movie clips. In summary, the spatial information of EEG is beneficial to discriminate emotional state.

In this work, we propose a transformer-based model, denoted as hierarchical spatial learning transformer (HSLT), to extract discriminative features by robustly capturing the EEG spatial dependencies from electrode level to brain-region level. The framework of HSLT include the following three parts:

(1) Division of the electrode patches. Inspired by the image spitting in the ViT [15], we treat PSD features from each electrode as the electrode patch. And we divide the electrode patches into different clusters according to the region classification of the cortex in the neuroscience.

(2) Electrode-level spatial learning. The electrode patches within different brain regions are separately inputted to the corresponding transformer encoders. It aims to integrate the critical information within the brain regions and lays a good foundation for learning the spatial information of overall brain.

(3) Brain-region-level spatial learning. The latent features from electrode-level transformer encoders are served as the brain region patches. And the patches are parallelly inputted to a transformer encoder to obtain the emotion prediction. The multi-head self-attention within the transformer encoder could enhance the capturing of spatial dependencies among the brain regions. Meanwhile, it could emphasize the contributive brain regions.

What's more, the learnable positional embedding and class token are adopted in all the transformer encoders of HSLT to learn essential positional information across the electrodes or brain regions, and aggregate representative information respectively.

The remainder of the paper is organized as follows. In section II, we introduce the related works. In section III, we describe the details of DEAP and MAHNOB-HCI database. In section IV, we specify the details of the HSLT for emotion recognition. In section V, we conduct extensive experiments to validate the effectiveness of HSLT. In section VI, we discuss the performance on different frequency bands, analyze the essential brain regions by performance evaluation and data visualization, compare the performance of HSLT with related works, and summarize the advantages and limitations of HSLT. Finally, we conclude this work in section VII.

II. Related Works

A. Spatial Information Learning for EEG-Based Emotion Recognition

Recently, many researchers apply the deep learning models to learn the spatial information of EEG. Khosrowabadi *et al.* [16] propose a biologically inspired feedforward neural network to learn the functional connectivity features between the brain regions. Although the effective spatial information could be found by this network, it is difficult for the feedforward neural network to learn the complex connectivity through a number of electrodes. Bashivan *et al.* [17] utilize CNN to robustly learn the spatial information of EEG using generated 2D EEG images according to the topology of the electrodes. Zhang *et al.* [18] propose a Recurrent Neural Network (RNN)-based model to capture the spatial dependencies among the electrodes. However, both convolutional and recurrent operation focus on the local neighborhood in space or time [19]. These will make the CNNs and RNNs have difficulty in capturing the long-range EEG spatial dependencies which is beneficial to extract critical information through a number of electrodes and brain regions.

B. Transformer and Its Application in Different EEG Tasks

To enhance the long-range dependencies capturing in EEG, attention mechanism based method [20], [21] have been adopted for improving the emotion recognition performance. As the self-attention based deep learning model, transformer [22] has attracted more and more attention. Transformer-based models have achieved the state-of-the-art performance in the fields of NLP and CV, such as the BERT [23] and ViT [15]. The multi-head self-attention and parallel inputting make the transformers have superior abilities to capture the long-range dependencies. Besides, the positional embedding retains the critical positional information of words and image patches, and class token could aggregate representative information. These strategies could potentially improve the performance of sequence analysis [15].

Recently, transformer-based models also have been proposed for different EEG tasks. Sun *et al.* [24] combine the transformer with CNN to improve the accuracy of motor imagery. Pedoeem *et al.* [25] build a hybrid architecture of convolutional layers, fully connected layers, and a transformer for seizure detection. These researches indicate that transformer is also a powerful model for extracting discriminative EEG features.

III. The Description of the Emotion Databases

In this work, the DEAP database [26] and MAHNOB-HCI [27] are chosen as the benchmark emotion databases. Here, we briefly introduce the experimental protocol and preprocessing of the EEG data.

The DEAP is a multimodal emotion database which includes EEG signals and peripheral signals from 32 subjects. The 40 one-minute-long music videos are utilized to elicit the emotions, and these videos are presented in 40 trails. In each trail, the EEG signals and peripheral signals are simultaneously recorded when subjects watching the videos. At the end of each trail, the subjects are rated the arousal, valence, and dominance according to Self-Assessment Manikin (SAM) with a range of 1 to 9.

The MAHNOB-HCI database is built for analysis emotions and implicit tagging. The experimental protocol of MAHNOB-HCI is similar to the DEAP. 20 movie clips are presented which are ranged from 34.9 seconds to 117 seconds long. During each trail, EEG signals and peripheral signals are recorded from 27 subjects, and subjects accomplish the self-report using arousal, valence, dominance, predictability and

其中这些脑区可以提供比其他脑区更多的贡献信息。此外，Sangnark 等人[14]和 Lakhan 等人[5]通过分析 EEG 对流行音乐和电影片段的响应，发现贡献性脑区或电极可以实现比其他更好的性能。总之，EEG 的空间信息有助于区分情绪状态。

在这项工作中，我们提出了一种基于 transformer 的模型，称为分层空间学习 transformer（HSLT），通过稳健地捕获从电极级别到脑区级别的 EEG 空间依赖性来提取判别性特征。HSLT 的框架包括以下三个部分：

(1) 电极块的划分。受 ViT[15]中图像分割的启发，我们将每个电极的 PSD 特征视为电极块。并根据神经科学中皮层区域分类，将电极块划分为不同的簇。

(2) 电极级空间学习。不同脑区的电极片被分别输入到相应的 Transformer 编码器中。其目的是整合脑区内的关键信息，为学习整体脑的空间信息打下良好基础。

(3) 脑区级空间学习。电极级 Transformer 编码器的潜在特征被作为脑区片。这些片并行输入到 Transformer 编码器中，以获得情绪预测。Transformer 编码器内的多头自注意力机制能够增强捕捉脑区间空间依赖关系的能力，同时能够强调贡献性脑区。

此外，HSLT 中的所有 Transformer 编码器都采用了可学习的位置嵌入和类别标记，以学习电极或脑区间的关键位置信息，并分别聚合代表性信息。

本文的其余部分组织如下。在第二节，我们介绍了相关工作。在第三节，我们描述了 DEAP 和 MAHNOB-HCI 数据库的详细信息。在第四节，我们详细说明了用于情绪识别的 HSLT。在第五节，我们进行了广泛的实验以验证 HSLT 的有效性。在第六节，我们讨论了不同频段上的性能，通过性能评估和数据可视化分析关键脑区，比较 HSLT 与相关工作的性能，并总结 HSLT 的优势和局限性。最后，我们在第七节总结这项工作。

II. 相关工作 A. 基于脑电的情绪识别的空间信息学习

最近，许多研究人员将深度学习模型应用于学习脑电的空间信息。Khosrowabadi 等人[16]提出了一种受生物物学启发的前馈神经网络，用于学习脑区之间的功能连接特征。尽管该网络可以通过这种方式找到有效的空间信息

对于前馈神经网络来说，通过多个电极学习复杂的连接性是困难的。Bashivan 等人[17]利用 CNN 根据电极的拓扑结构生成 2D EEG 图像，以稳健地学习 EEG 的空间信息。Zhang 等人[18]提出了一种基于循环神经网络（RNN）的模型来捕获电极之间的空间依赖关系。然而，卷积操作和循环操作都集中在空间或时间上的局部邻域[19]。这将使 CNN 和 RNN 难以捕获长程 EEG 空间依赖关系，而这种依赖关系有利于通过多个电极和大脑区域提取关键信息。

B. Transformer 及其在不同 EEG 任务中的应用

为了增强脑电图（EEG）中长距离依赖关系的捕捉，我们采用了基于注意力机制的方法[20]、[21]来提高情绪识别性能。作为基于自注意力机制的深度学习模型，Transformer[22]已引起越来越多的关注。基于 Transformer 的模型在自然语言处理（NLP）和计算机视觉（CV）领域取得了最先进的性能，例如 BERT[23]和 ViT[15]。多头自注意力机制和并行输入使 Transformer 具有捕捉长距离依赖关系的优越能力。此外，位置嵌入保留了单词和图像块的关键位置信息，而类别标记可以聚合代表性信息。这些策略有可能提高序列分析的性能[15]。

最近，基于 transformer 的模型也被提出用于不同的脑电图（EEG）任务。Sun 等人[24]将 transformer 与卷积神经网络（CNN）结合以提高运动想象的识别准确率。Pedoeem 等人[25]构建了一个混合架构，包含卷积层、全连接层和 transformer，用于癫痫检测。这些研究表明，transformer 也是一个强大的模型，用于提取具有判别性的 EEG 特征。

III. T D ESCRIPTION OF THE E D

在这项工作中，DEAP 数据库[26]和 MAHNOBHCI[27]被选为基准情绪数据库。这里，我们简要介绍实验协议和 EEG 数据的预处理。

DEAP 是一个多模态情绪数据库，包括来自 32 名受试者的脑电图（EEG）信号和外周信号。利用 40 个一分钟的音乐视频来诱发情绪，这些视频以 40 个试验的顺序呈现。在每个试验中，当受试者观看视频时，同时记录脑电图信号和外周信号。在每个试验结束时，受试者根据自我评估娃娃（SAM）对唤醒度、效价和支配性进行评分，评分范围为 1 到 9。

MAHNOB-HCI 数据库用于分析情绪和隐式标记。MAHNOB-HCI 的实验协议与 DEAP 相似。展示了 20 个电影片段，时长从 34.9 秒到 117 秒不等。在每个试验中，从 27 名受试者那里记录脑电图信号和外周信号，受试者使用唤醒度、效价、支配性和可预测性完成自我报告。

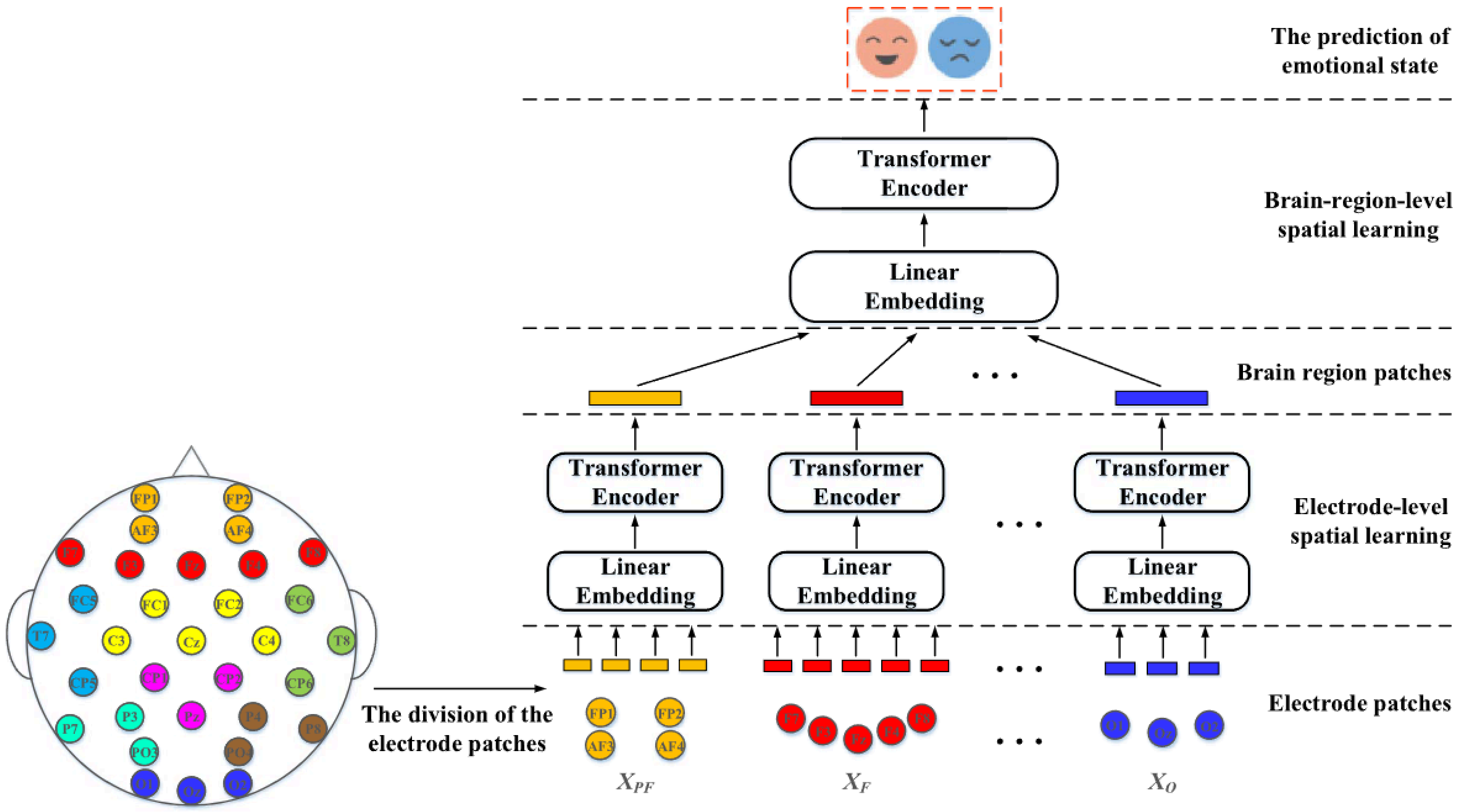

Fig. 1. The overview of the HSLT. We extract the PSD features from each electrode and treat the aggregation of PSD features from each electrode as the electrode patch. Next, electrode patches are divided according to the layout of the brain regions, the electrodes with the same color are divided into the same brain region. And the EEG spatial information learning includes two stages: the electrode-level spatial learning and the brain-region-level spatial learning.

emotional keywords. The EEG acquisition equipment is also the 32-channel Biosemi system, so the preprocessing of EEG data is consistent with the DEAP. But the EEG data of 24 subjects are available due to the incomplete data of 3 subjects.

The EEG signals of two datasets are both recorded by a 32-channel Biosemi system with a sampling rate of 512 Hz, and the electrodes are placed according to the 10-20 system. Firstly, the EEG signals are downsampled to 128 Hz. Next, a 4-45 Hz bandpass filter is applied. Finally, the electrooculography (EOG) artifacts are removed by the blind source separation technique using EEGLAB toolbox [28].

IV. The Novel HSLT for EEG Emotion Recognition

A. The Overview of the HSLT

In this section, we will identify the configuration of the HSLT. As shown in Fig. 1., HSLT includes three major parts: EEG feature extraction and division of the electrode patches, electrode-level spatial learning and brain-region-level spatial learning. We will briefly introduce these three parts in the remainder of this section.

B. EEG Feature Extraction and Division of the Electrode Patches

For each EEG segment, we extract the PSD features from different frequency bands. What's more, the PSD features are obtained by the Welch method with a one-second-long Hamming window. The aggregation of PSD features from one electrode, denoted as the electrode patch, can be formulated as $x_i \in \mathbb{R}^d$, where i denotes the i-th electrode and d is number of the frequency bands. And the EEG feature set can be denoted as $X = [x_1, x_2, \ldots, x_e] \in \mathbb{R}^{e\times d}$, where e is the number of electrodes.

TABLE I
The Electrode Patch Clusters Associated With Different Brain Regions

Brain region	Electrodes	Representation of the clusters
Pre-frontal (PF)	FP1, AF3, AF4, FP2	$X_{PF} \in \mathbb{R}^{4\times d}$
Frontal (F)	F7, F3, Fz, F4, F8	$X_F \in \mathbb{R}^{5\times d}$
Left temporal (LT)	FC5, T7, CP5	$X_{LT} \in \mathbb{R}^{3\times d}$
Central (C)	FC1, C3, Cz, C4, FC2	$X_C \in \mathbb{R}^{5\times d}$
Right Temporal (RT)	FC6, T8, CP6	$X_{RT} \in \mathbb{R}^{3\times d}$
Left Parietal (LP)	P7, P3, PO3	$X_{LP} \in \mathbb{R}^{3\times d}$
Parietal (P)	CP1, Pz, CP2	$X_P \in \mathbb{R}^{3\times d}$
Right Parietal (RP)	P8, P4, PO4	$X_{RP} \in \mathbb{R}^{3\times d}$
Occipital (O)	O1, Oz, O2	$X_O \in \mathbb{R}^{3\times d}$

The brain response to different emotions could be reflected by the activations of various brain regions [29], [30]. Hence, the differences among the activations of the brain regions could supply the informative feature to distinguish the emotions. The lobes of the brain are the anatomical classification of the cortex, the frontal lobe, central lobe, temporal lobe, parietal lobe and occipital lobe are related to different brain functions [31]. According to the criterion, we split the electrode patches into different clusters to represent the functions of various brain regions. As shown in the Fig. 1. and Table I, the pre-frontal, frontal, and so on are the nine clusters utilized in this work.

C. The Details of the Transformer Encoder in the HSLT

The overview of the transformer encoder in the HSLT is shown in Fig. 2. Given the representation of electrode patches

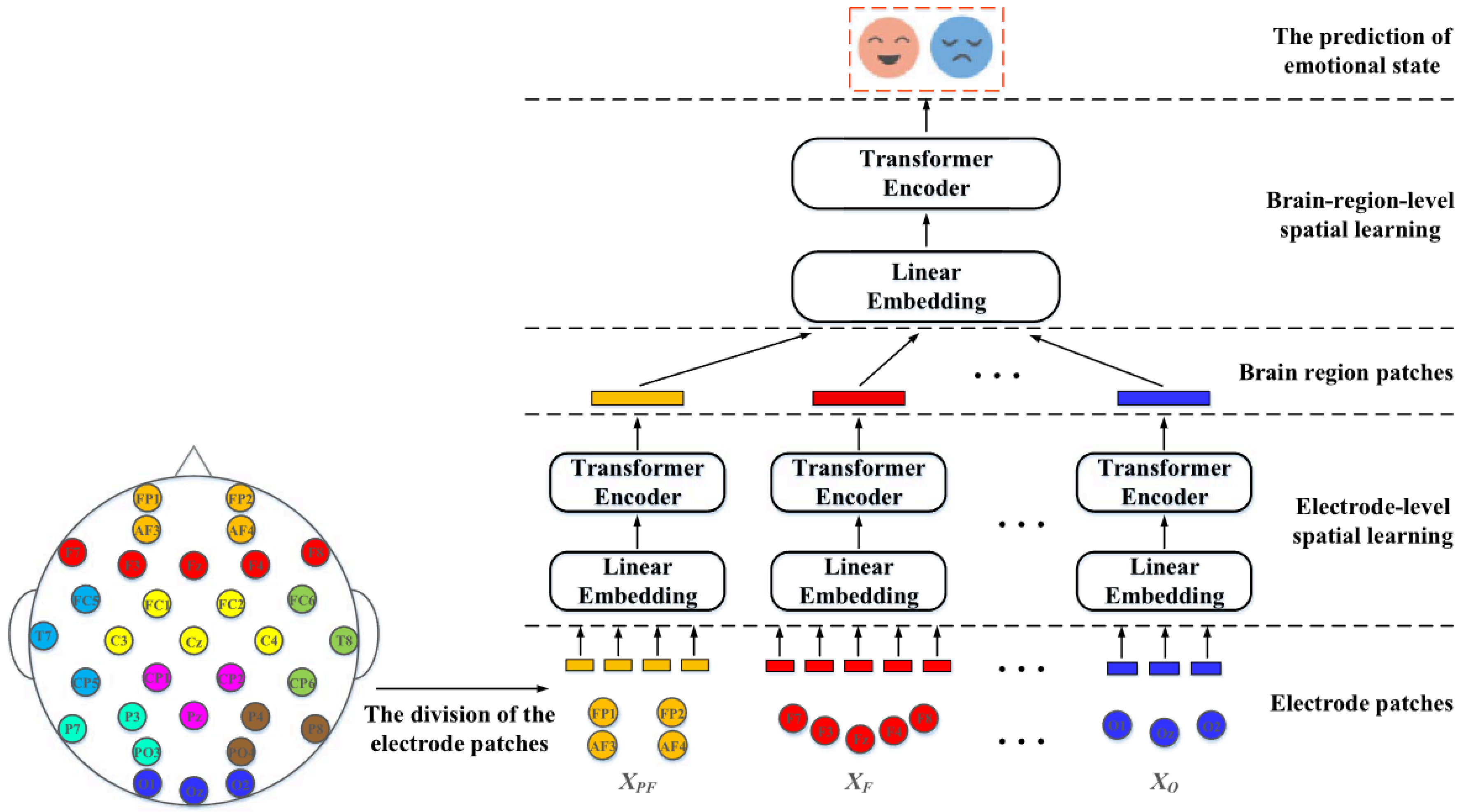

图 1. HSLT 的概述。我们从每个电极提取 PSD 特征，并将每个电极的 PSD 特征聚合视为电极补丁。接下来，根据大脑区域的布局将电极补丁进行划分，颜色相同的电极被划分为同一脑区。EEG 空间信息学习包括两个阶段：电极级空间学习和脑区级空间学习。

情绪关键词。EEG 采集设备也是 32 通道的 Biosemi 系统，因此 EEG 数据的预处理与 DEAP 一致。但由于 3 名受试者的数据不完整，只有 24 名受试者的 EEG 数据可用。

两个数据集的 EEG 信号均由采样率为 512 Hz 的 32 通道 Biosemi 系统记录，电极按照 10-20 系统放置。首先，将 EEG 信号下采样至 128 Hz。接下来，应用 4-45 Hz 带通滤波器。最后，使用 EEGLAB 工具箱[28]中的盲源分离技术去除眼动伪迹。

IV. T N HSLT EEG

E R A. HSLT 概述

在本节中，我们将确定 HSLT 的配置。如图 1 所示，HSLT 包括三个主要部分：脑电图特征提取和电极块的划分、电极级空间学习和脑区级空间学习。我们将在本节剩余部分简要介绍这三个部分。

B. 脑电图特征提取和电极块划分

对于每个脑电图（EEG）片段，我们从不同的频段提取功率谱密度（PSD）特征。此外，PSD 特征是通过使用一秒钟长的汉明窗的 Welch 方法获得的。一个电极的 PSD 特征的聚合，记为电极贴片，可以表示为 x ∈ R，其中 i 表示第 i 个电极，d 是频段的数量。EEG 特征集可以表示为

表 I

T E PC A

W D B R

Brain region	Electrodes	Representation of the clusters
Pre-frontal (PF)	FP1, AF3, AF4, FP2	$X_{PF} \in \mathbb{R}^{4\times d}$
Frontal (F)	F7, F3, Fz, F4, F8	$X_F \in \mathbb{R}^{5\times d}$
Left temporal (LT)	FC5, T7, CP5	$X_{LT} \in \mathbb{R}^{3\times d}$
Central (C)	FC1, C3, Cz, C4, FC2	$X_C \in \mathbb{R}^{5\times d}$
Right Temporal (RT)	FC6, T8, CP6	$X_{RT} \in \mathbb{R}^{3\times d}$
Left Parietal (LP)	P7, P3, PO3	$X_{LP} \in \mathbb{R}^{3\times d}$
Parietal (P)	CP1, Pz, CP2	$X_P \in \mathbb{R}^{3\times d}$
Right Parietal (RP)	P8, P4, PO4	$X_{RP} \in \mathbb{R}^{3\times d}$
Occipital (O)	O1, Oz, O2	$X_O \in \mathbb{R}^{3\times d}$

X = [x, x, . . . , x] ∈ R，其中 e 是电极的数量。

大脑对不同情绪的反应可以通过不同脑区的激活来反映[29], [30]。因此，脑区激活的差异可以提供区分情绪的信息特征。大脑的叶是皮层的解剖分类，额叶、中央叶、颞叶、顶叶和枕叶与不同的脑功能相关[31]。根据这一标准，我们将电极片划分为不同的簇以代表不同脑区的功能。如图 1 和表 I 所示，本工作使用了九个簇，包括前额叶、额叶等。

C. HSLT 中 Transformer 编码器的细节

HSLT 中 transformer 编码器的概述如图 2 所示。考虑到电极块的表示

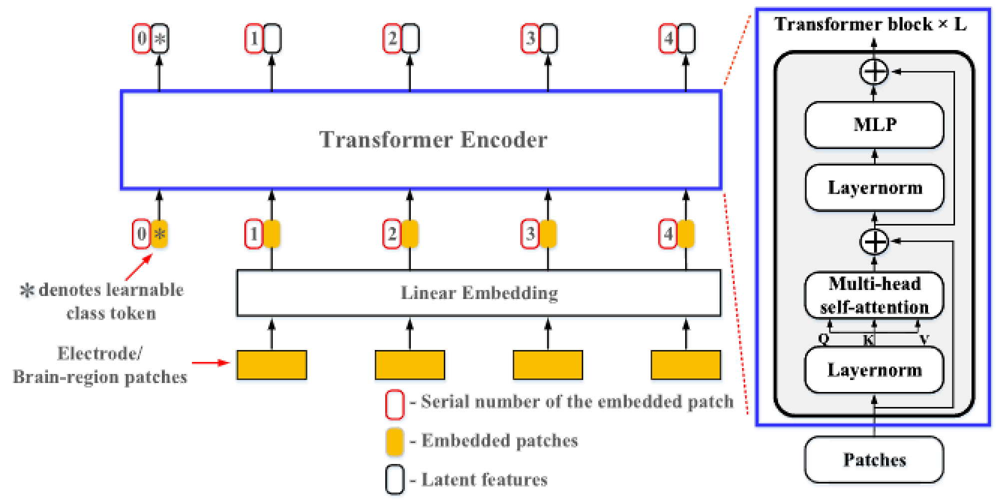

Fig. 2. The overview of the transformer encoder in the HSLT. A transformer encoder contains L stacked transformer blocks.

within the brain region $X_E = [X_E^1, X_E^2, \ldots, X_E^N] \in \mathbb{R}^{N\times d}$. Firstly, the electrode patches are mapped to a constant size D_e using linear embedding. According to the Eq.1, we could obtain the representation of patch embeddings $E \in \mathbb{R}^{(N+1)\times D_e}$, where $x_E^{cls} \in \mathbb{R}^{D_e}$ denotes the class token in the electrode-level learning. The class token is an extra learnable embedding which could aggregate representative information of all embeddings [18]. Besides, $W_E \in \mathbb{R}^{d\times D_e}$ is the linear projection matrix.

$$E = \left[x_E^{cls}; x_E^1 W_E; x_E^2 W_E; \ldots; x_E^N W_E\right] \tag{1}$$

Next, we utilized the 1-D positional embedding $E_E^{pos} \in \mathbb{R}^{(N+1)\times D_e}$ which aims to retain the spatial information of the electrode patches or region patches. According to the Eq.2, the result Z_E is served as input of the transformer encoder.

$$Z_E = E + E_E^{pos} \tag{2}$$

As shown in Fig.2, the transformer block includes multi-head self-attention, layer normalization and Multiple Layer Perception (MLP). The Multi-head Self-Attention (MSA) is an extension of self-attention which is widely used in NLP [32]. And the Layer Normalization (LN) could reduce the training time and improve the generalization performance [33]. Here, we given L_e as the block number in the electrode-level spatial learning. The operation in the transformer encoder is shown as Eq.3 and Eq.4.

$$Z'_{l_e} = MSA\left(LN\left(Z_{l_e-1}\right)\right) + Z_{l_e-1} \quad l = 1, \ldots, L \tag{3}$$

$$Z_{l_e} = MLP\left(LN\left(Z'_{l_e}\right)\right) + Z'_{l_e} \quad l = 1, \ldots, L \tag{4}$$

where $MSA(\cdot)$ and $MLP(\cdot)$ denote MSA operation and MLP operation, Z'_{l_e} and Z_{l_e} are the output of the MSA and MLP respectively. The L is number of the stacked transformer blocks, so the output of the last block is $Z_{L_e} \in \mathbb{R}^{(N+1)\times D_e}$.

In the electrode-level spatial learning, the electrode patches within one brain region are parallelly inputted to the corresponding transformer encoder. According to the division of electrodes in the Table I, there are nine transformer encoders which are corresponding to the brain regions. The latent features X_L obtained by nine transformer encoders could be formulated as:

$$X_L = \left[Z_{L_e}^{PF}; Z_{L_e}^{F}; Z_{L_e}^{LT}; Z_{L_e}^{C}; Z_{L_e}^{RT}; Z_{L_e}^{LP}; Z_{L_e}^{P}; Z_{L_e}^{RP}; Z_{L_e}^{o}\right] \tag{5}$$

The dimensionality of each element in the X_L is not equal. The reason is that the PF, F, and C brain regions include more than three electrodes, while other brain regions only contain three electrodes. To guarantee the same dimensionality, $Z_{L_e}^{PF}$, $Z_{L_e}^{F}$, and $Z_{L_e}^{C}$ are projected into the $4 \times D_e$ dimensions. According to the Eq.6, we obtained the patches $X_R \in \mathbb{R}^{9\times 4\times D_e}$ for brain-region-level spatial learning.

$$\begin{aligned} X_R &= \left[Z_{L_e}^{PF} W_{PF}; Z_{L_e}^{F} W_F; Z_{L_e}^{LT}; Z_{L_e}^{C} W_C; Z_{L_e}^{RT}; Z_{L_e}^{LP}; Z_{L_e}^{P};\right. \\ &\quad \left. Z_{L_e}^{RP}; Z_{L_e}^{o}\right] \\ &= [X_R^1, X_R^2, \ldots, X_R^9] \in \mathbb{R}^{9\times 4\times D_e} \end{aligned} \tag{6}$$

In the brain-region spatial learning, the operations are similar to the electrode-level spatial learning, and the operations can be represented as:

$$Z_R = \left[x_R^{class}; x_R^1 W_R; x_R^2 W_R; \ldots; x_R^9 W_R\right] + E_R^{pos} \tag{7}$$

$$Z'_{l_r} = MSA\left(LN\left(Z_{l_r-1}\right)\right) + Z_{l_r-1} \quad l_r = 1, \ldots, L_r \tag{8}$$

$$Z_{l_r} = MLP\left(LN\left(Z'_{l_r}\right)\right) + Z'_{l_r} \quad l_r = 1, \ldots, L_r \tag{9}$$

where the $W_R \in \mathbb{R}^{4\times D_e\times D_r}$ is the weight of the linear projection, $E_R^{pos} \in \mathbb{R}^{10\times D_r}$ is the positional embedding, Z'_{l_r} and Z_{l_r} are the outputs of the MSA and MLP respectively. The output $Z_{L_r}^0 \in \mathbb{R}^{D_R}$ is corresponding to the x_R^{class}. Finally, the prediction of arousal or valence is obtained by Eq.10.

$$\hat{y} = \sigma(W_O Z_{L_R}^0) \tag{10}$$

where $\hat{y}$ is the prediction result, W_O is the weight, and $\sigma(\cdot)$ is the sigmoid function.

V. Results

In this section, we separately conduct the subject-independent experiments on the DEAP and MAHNOB-HCI database to validate the effectiveness of HSLT. In each subject-independent experiment, the leave-one-subject-out cross-validation are adopted to derive the classification performance. Specifically, EEG data from one subject is considered as the testing set, and EEG data from the remaining subjects are set as training set until each subject's data has been set as test set once. And the final performance is the average of all the folds.

The sufficient data are necessary to provide for training transformer-based model [15]. Hence, we adopt a 6-second-long sliding window with 50% overlap to segment the EEG data. In this work, the self-assessment of 1-4 is treated as 'low class,' the low arousal (LA) and low valence (LV). And the self-assessment of 6-9 is treated as 'high class,' the high arousal (HA) and high valence (HV). According to the criterion, we transform the emotion recognition problem into the binary classification (LA vs. HA, and LV vs. HV) and the four-classes classification (LALV vs. LAHV vs. HALV vs. HAHV). What's more, to deal with the imbalance samples problem in the two databases shown in Table II and Table III, we adopt the accuracy (denoted as P_{acc}), weighted F1-score (denoted as P_f), and Cohen's kappa coefficient (denoted as P_{ck}) to evaluate the classification performance.

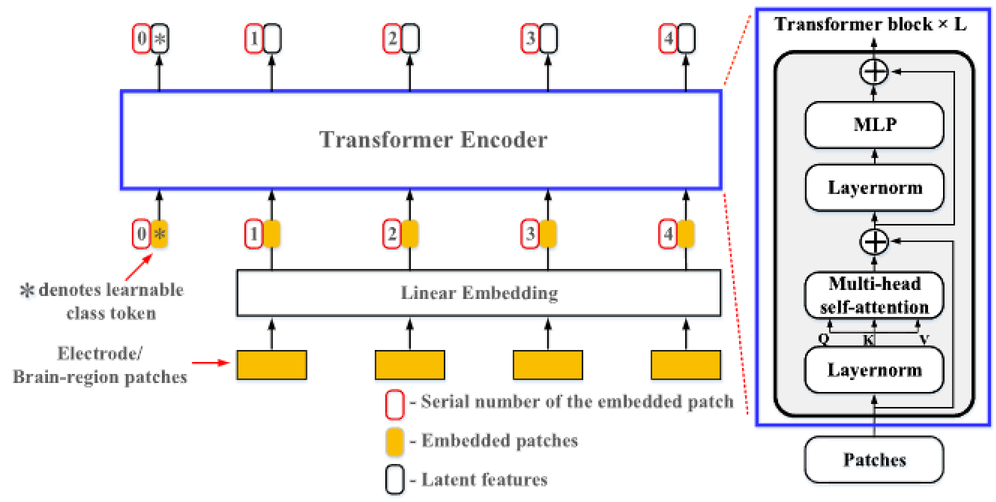

图 2. HSLT 中的 Transformer 编码器概述。Transformer 编码器包含 L 个堆叠的 Transformer 模块。

在脑区 X = [X, X, . . . , X]∈ R 内。首先，电极贴片通过线性嵌入映射到固定大小 D。根据式 1，我们可以得到贴片嵌入的表示 E∈ R，其中 x ∈ R 表示电极级别学习中的类标记。类标记是一个额外的可学习嵌入，可以聚合所有嵌入的代表性信息[18]。此外，W ∈ R 是线性投影矩阵。

$$E = [x ; xW ; xW ; . . . ; x W] \quad (1)$$

接下来，我们使用了 1-D 位置嵌入 E E ∈ R，旨在保留电极块或区域块的 spatial 信息。根据 Eq.2，结果 Z 作为 transformer 编码器的输入。

$$Z = E + E E \quad (2)$$

如图 2 所示，transformer 模块包括多头自注意力、层归一化和多层感知机 (MLP)。多头自注意力 (MSA) 是自注意力的扩展，在自然语言处理 (NLP) 中被广泛使用 [32]。层归一化 (LN) 可以减少训练时间并提高泛化性能 [33]。这里，我们将 L 作为电极级空间学习中的模块数量。transformer 编码器中的操作显示为 Eq.3 和 Eq.4。

$$Z = MSA(LN(Z)) + Z \quad l = 1, . . . , L \quad (3)$$

$$Z = MLP(LN(Z)) + Z \quad l = 1, . . . , L \quad (4)$$

其中 M S A(·) 和 M L P(·) 分别表示 MSA 操作和 MLP 操作，Z 和 Z 分别是 MSA 和 MLP 的输出。L 是堆叠的 transformer 块的数量，因此最后一个块的输出是 Z ∈ R。

在电极级别的空间学习中，一个脑区的电极块被并行输入到相应的 Transformer 编码器中。根据表 I 中电极的划分，有九个 Transformer 编码器对应于不同的脑区。九个 Transformer 编码器获得的潜在特征 X 可以表示为:

$$X = [Z L ; Z ; Z ; Z ; Z ; Z L ; Z ; Z L ; Z] \quad (5)$$

X 中的每个元素的维度并不相同。

原因是 PF、F 和 C 脑区包含超过三个电极，而其他脑区只包含三个电极。为了保证相同的维度，Z L 、Z 和 Z 被投影到 4 × D 维度。根据公式 6，我们获得了用于脑区级空间学习的 X ∈ R 的块。

$$X = [Z L W ; Z W ; Z ; Z W; Z L ; Z ; Z ; Z L ; Z] = [X, X, . . . , X] \in R \quad (6)$$

在大脑区域空间学习中，操作与电极级空间学习类似，操作可以表示为:

$$Z = [x R ; xW ; xW ; . . . ; xW] + ER \quad (7)$$

$$Z = MSA(LN(Z)) + Z \quad l = 1, . . . , L \quad (8)$$

$$Z = MLP(LN(Z)) + Z \quad l = 1, . . . , L \quad (9)$$

其中 W ∈ R 是线性投影的权重，E R ∈ R 是位置嵌入，Z 和 Z 分别是 MSA 和 MLP 的输出。输出 Z ∈ R 对应于 x R 。最后，通过公式 10 获得唤醒度或效价预测。

$$\hat{y} = \sigma(W Z) \quad (10)$$

其中 ˆy 是预测结果，W 是权重，σ(·) 是 sigmoid 函数。

V. R

在本节中，我们分别对 DEAP 和 MAHNOB-HCI 数据库进行独立于被试的实验，以验证 HSLT 的有效性。在每个独立于被试的实验中，采用留一被试交叉验证方法来获得分类性能。具体来说，一个被试的脑电数据被视为测试集，其余被试的脑电数据设置为训练集，直到每个被试的数据都作为测试集使用一次。最终性能是所有折数的平均值。

足够的数据对于训练基于 transformer 的模型是必要的[15]。因此，我们采用 6 秒长的滑动窗口，重叠率为 50%来分割脑电图数据。在这项工作中，自我评估 1-4 被处理为“低类别”，即低唤醒（LA）和低效价（LV）。而自我评估 6-9 被处理为“高类别”，即高唤醒（HA）和高效价（HV）。根据标准，我们将情绪识别问题转化为二元分类（LA 与 HA，以及 LV 与 HV）和四类分类（LALV 与 LAHV 与 HALV 与 HAHV）。此外，为了处理表 II 和表 III 中两个数据库中的样本不平衡问题，我们采用准确率（记为 P）、加权 F1 分数（记为 P）和 Cohen's kappa 系数（记为 P）来评估分类性能。

TABLE II
THE NUMBER OF SAMPLES IN THE BINARY CLASSIFICATION

Database	LA	HA	LV	HV
DEAP	7144	10108	6973	10374
MAHNOB -HCI	8538	6585	8588	6241

TABLE III
THE NUMBER OF SAMPLES IN THE FOUR-CLASS CLASSIFICATION

Database	LALV	LAHV	HALV	HAHV
DEAP	2413	2014	2983	5225
MAHNOB -HCI	3097	2972	4118	2151

TABLE IV
THE RESULTS OF HSLT IN THE BINARY CLASSIFICATION

Database	LA vs. HA			LV vs. HV		
	P_{acc} (%)	P_f (%)	P_{ck}	P_{acc} (%)	P_f (%)	P_{ck}
DEAP	65.75 (8.51)	64.29 (10.06)	0.439 (0.112)	66.51 (8.53)	66.27 (7.29)	0.445 (0.109)
MAHNOB -HCI	66.20 (10.48)	63.85 (10.23)	0.457 (0.129)	66.63 (6.51)	65.33 (7.21)	0.463 (0.101)

The standard deviations are in the brackets.

TABLE V
THE RESULTS OF HSLT IN THE FOUR-CLASS CLASSIFICATION

Database	P_{acc} (%)	P_f (%)	P_{ck}
DEAP	56.93 (8.22)	54.29 (8.59)	0.327 (0.094)
MAHNOB-HCI	58.03 (9.36)	55.32 (9.71)	0.343 (0.107)

TABLE VI
THE RESULTS OF DEEP NETWORKS IN THE DEAP DATABASE (%)

	LA vs. HA		LV vs. HV	
	P_{acc}	P_f	P_{acc}	P_f
The proposed HSLT	**65.75 (8.51)**	64.29 (10.06)	**66.51 (8.53)**	66.27 (7.29)
One-stage Transformer#	61.47 (11.83)	59.44 (11.82)	61.89 (8.67)	59.65 (9.25)
CNN* [34]	60.72 (10.10)	58.88 (10.83)	61.43 (8.60)	60.68 (8.08)
LSTM* [35]	58.39 (9.04)	52.61 (9.85)	58.65 (8.16)	53.31 (8.90)
DBN* [36]	59.82 (8.72)	57.49 (8.23)	60.70 (8.93)	59.72 (8.41)
CNN-LSTM* [37]	62.10 (9.63)	60.47 (9.99)	63.86 (8.55)	61.36 (7.73)
DenseNet* [38]	63.37 (7.05)	61.50 (7.47)	64.57 (7.48)	62.58 (7.06)

A. The Classification Results of HSLT

The PSD features are extracted on five bands, Theta band (4-7 Hz), Slow alpha band (8-10 Hz), Alpha band (8-12 Hz), Beta band (13-30 Hz) and Gamma band (30-47 Hz). Hence, the dimension of the feature d is 5. We use the dropout with 0.4 in all the MSA modules and MLP modules. In addition, we use the dropout with 0.1 in the linear patch projections of brain-region-level learning. For training the HSLT, the Adam optimizer with cosine learning decay is adopted, batch size is set as 512, and epoch is 80 with early stopping. The loss function is cross entropy function, and labels are processed according to the one-hot coding. All the networks in this work are implemented by pytorch with two NVIDIA GeForce RTX 2080 GPUs. Additionally, we set the hyperparameter L_e, L_r both as 2, D_e, D_r are fixed as 8 and 16 respectively. MSA is adopted in the electrode-level and brain-region-level, D_h and k are set as 64 and 16 respectively.

The classification results of binary and four-class classification in two databases are listed in Table IV and Table V. In the binary classification, the HSLT achieves the accuracies of 65.75% and 66.51% with arousal and valence level in the DEAP database. Meanwhile, HSLT also obtain the accuracies of 66.20% and 66.63% in the MAHNOB-HCI database. Compared with binary classification, the performance of four-class classification properly declines due to the increasing difficulty of classification. The HSLT achieves the accuracies of 56.93% and 58.03% with arousal and valence level in the DEAP and MAHNOB-HCI database, respectively.

B. Comparison With Other Deep Networks

The results of different methods in two databases have been illustrated in Table VI and Table VII. According to the results, the proposed HSLT has achieved optimal performance and it out- performs the other networks in both two databases. In contrast to the accuracy of CNN, LSTM and DBN, the accuracy of proposed HSLT have been boosted more than 5% and 6% in the DEAP and MAHNOB-HCI database respectively. Moreover, the proposed HSLT surpasses the one-stage Transformer over 4%. Similarly, the one-stage Transformer is also superior to these three networks. In addition, we also reimplement the latest spatial and temporal EEG encoding networks, CNN-LSTM and DenseNet. Compared with these two networks, the HSLT also achieves a slight improvement.

Except for the proposed HSLT and one-stage transformer, the attention mechanisms are not adopted in the other networks. According to the comparison, the self-attention within HSLT could enhance the long-range dependencies capturing in EEG. Besides, the standard deviation of HSLT is basically less than the other networks. It suggests that more stable performances have achieved by HSLT in the test set. In summary, these results could validate the effectiveness of the proposed HSLT in the emotion recognition.

表 II

T N S B C

Database	LA	HA	LV	HV
DEAP	7144	10108	6973	10374
MAHNOB -HCI	8538	6585	8588	6241

表 III

T N S F -C C

Database	LALV	LAHV	HALV	HAHV
DEAP	2413	2014	2983	5225
MAHNOB -HCI	3097	2972	4118	2151

表 IV

T R HSLT B C

Database	LA vs. HA			LV vs. HV		
	P_{acc} (%)	P_f (%)	P_{ck}	P_{acc} (%)	P_f (%)	P_{ck}
DEAP	65.75 (8.51)	64.29 (10.06)	0.439 (0.112)	66.51 (8.53)	66.27 (7.29)	0.445 (0.109)
MAHNOB -HCI	66.20 (10.48)	63.85 (10.23)	0.457 (0.129)	66.63 (6.51)	65.33 (7.21)	0.463 (0.101)

The standard deviations are in the brackets.

A. HSLT 的分类结果

PSD 特征在五个频段上提取，包括 Theta 频段（4-7 Hz）、慢 Alpha 频段（8-10 Hz）、Alpha 频段（8-12 Hz）、Beta 频段（13-30 Hz）和 Gamma 频段（30-47 Hz）。因此，特征维度 d 为 5。我们在所有 MSA 模块和 MLP 模块中使用 0.4 的 dropout。此外，在脑区级别的线性补丁投影中使用 0.1 的 dropout。为了训练 HSLT，我们采用带有余弦学习衰减的 Adam 优化器，批大小设置为 512，训练轮数为 80，并使用提前停止。损失函数为交叉熵函数，标签根据 one-hot 编码处理。本工作中的所有网络均使用 pytorch 实现，并配备两块 NVIDIA GeForce RTX 2080 GPU。此外，我们设置超参数 L 和 L 为 2，D 和 D 分别固定为 8 和 16。在电极级别和脑区级别采用 MSA，D 和 k 分别设置为 64 和 16。

二元分类和四分类在两个数据库中的分类结果分别列于表 IV 和表 V。在二元分类中，HSLT 在 DEAP 数据库的唤醒度和效价水平上分别达到了 65.75%和 66.51%的准确率。同时，HSLT 在 MAHNOB-HCI 数据库中也获得了 66.20%和 66.63%的准确率。与二元分类相比，四分类的性能略有下降，因为难度有所增加。

表 V

T R HSLT F -C C

Database	P_{acc} (%)	P_f (%)	P_{ck}
DEAP	56.93 (8.22)	54.29 (8.59)	0.327 (0.094)
MAHNOB-HCI	58.03 (9.36)	55.32 (9.71)	0.343 (0.107)

表 VI

T R D N DEAP D (%)

	LA vs. HA		LV vs. HV	
	P_{acc}	P_f	P_{acc}	P_f
The proposed HSLT	**65.75 (8.51)**	64.29 (10.06)	**66.51 (8.53)**	66.27 (7.29)
One-stage Transformer#	61.47 (11.83)	59.44 (11.82)	61.89 (8.67)	59.65 (9.25)
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CNN-LSTM* [37]	62.10 (9.63)	60.47 (9.99)	63.86 (8.55)	61.36 (7.73)
DenseNet* [38]	63.37 (7.05)	61.50 (7.47)	64.57 (7.48)	62.58 (7.06)

在分类方面。HSLT 在 DEAP 和 MAHNOB-HCI 数据库中，分别以唤醒度和效价水平达到了 56.93%和 58.03%的准确率。

B. 与其他深度网络的比较

不同方法在两个数据库中的结果已分别展示在表 VI 和表 VII 中。根据结果，所提出的 HSLT 实现了最佳性能，并且在两个数据库中均优于其他网络。与 CNN、LSTM 和 DBN 的准确率相比，所提出的 HSLT 在 DEAP 和 MAHNOB-HCI 数据库中的准确率分别提升了 5%和 6%。此外，所提出的 HSLT 还比单阶段 Transformer 高出 4%以上。类似地，单阶段 Transformer 也优于这三个网络。此外，我们还重新实现了最新的时空 EEG 编码网络 CNN-LSTM 和 DenseNet。与这两个网络相比，HSLT 也实现了小幅提升。

除了所提出的 HSLT 和单阶段 Transformer 外，其他网络并未采用注意力机制。根据比较，HSLT 中的自注意力机制能够增强在脑电图（EEG）中捕获的长程依赖关系。此外，HSLT 的标准差基本上小于其他网络。这表明 HSLT 在测试集上实现了更稳定的性能。总之，这些结果可以验证所提出的 HSLT 在情绪识别中的有效性。

TABLE VII
THE RESULTS OF DEEP NETWORKS IN THE MAHNOB-HCI DATABASE (%)

	LA vs. HA		LV vs. HV	
	P_{acc}	P_f	P_{acc}	P_f
The proposed HSLT	**66.20 (10.48)**	63.85 (10.23)	**66.63 (6.51)**	65.53 (7.21)
One-stage Transformer#	61.34 (10.85)	58.82 (10.02)	61.51 (7.17)	60.21 (7.20)
CNN* [34]	58.15 (10.62)	54.71 (10.54)	59.47 (5.96)	58.92 (6.62)
LSTM* [35]	56.84 (10.28)	53.87 (9.36)	57.98 (6.89)	53.67 (7.06)
DBN* [36]	56.95 (10.93)	53.31 (10.02)	59.78 (9.36)	58.59 (9.49)
CNN-LSTM* [37]	62.31 (9.07)	60.70 (8.53)	62.74 (8.21)	61.05 (8.69)
DenseNet* [38]	63.85 (9.15)	61.52 (8.76)	64.10 (8.62)	61.90 (7.99)

*The model with * denotes that we conduct same experiment by our own reimplementation.*
One-stage Transformer # means the all the 32 electrode patches are parallelly inputted into a transformer encoder. And the hyperparameters of one-stage HSLT are same to the transformer encoder of proposed HSLT in the brain-region-level spatial learning.

TABLE VIII
THE HYPERPARAMETERS OF DIFFERENT HSLT CONFIGURATIONS

	L_e	L_R	D_e	D_R	D_h	k
HSLT-Small	1	1	8	16	32	8
HSLT-Base	1	2	8	16	32	12
The proposed HSLT	2	2	8	16	64	16
HSLT-Large	2	3	8	16	64	16

C. HSLT With Different Configurations

In this part, we compare the performances between the different configurations of HSLT. Hence, we propose the four candidate configurations of HSLT, as the parameters summarized in Table VIII The configurations are denoted as 'HSLT-small,' 'HSLT-base,' 'The proposed HSLT' and 'HSLT-Huge' with the depth and head increasing.

The performances of different HSLT configurations in the DEAP database are listed in Table IX. The proposed HSLT achieves the best performance in the arousal and valence classification, and the performance of HSLT-Large is similar to the proposed HSLT. Furthermore, compared with HSLT-small and HSLT-base, the proposed HSLT and HSLT-Large significantly improve the accuracy with the paired t-test results ($p < 0.05$). It indicates that more discriminative feature could be learned by HSLT with the depth and head increasing. Additionally, the accuracy of proposed HSLT is significant outperformed HSLT-Large in the valence classification. And the accuracy of the proposed HSLT improve 0.13% than the HSLT-Large in the arousal classification.

TABLE IX
THE RESULTS OF DIFFERENT HSLT CONFIGURATIONS IN THE DEAP DATABASE (%)

	LA vs. HA		LV vs. HV	
	P_{acc}	P_f	P_{acc}	P_f
HSLT-Small	63.07 (10.49)	62.98 (8.90)	64.03 (8.40)	63.69 (7.75)
HSLT-Base	64.29 (10.85)	63.88 (10.02)	64.47 (8.36)	63.98 (7.42)
The proposed HSLT	**65.75 (8.51)**	64.29 (10.06)	**66.51* (8.53)**	66.27 (7.29)
HSLT-Large	65.62 (9.29)	64.18 (9.72)	65.50 (8.85)	64.62 (8.91)

TABLE X
THE RESULTS OF DIFFERENT HSLT CONFIGURATIONS IN THE MAHNOB-HCI DATABASE (%)

	LA vs. HA		LV vs. HV	
	P_{acc}	P_f	P_{acc}	P_f
HSLT-Small	62.79 (10.72)	61.69 (10.37)	64.65 (6.69)	63.88 (8.60)
HSLT-Base	64.14 (11.79)	63.34 (12.32)	65.56 (6.74)	64.75 (8.23)
The proposed HSLT	**66.20* (10.48)**	63.85 (10.23)	**66.63 (6.51)**	65.53 (7.21)
HSLT-Large	64.01 (11.61)	62.91 (11.06)	66.29 (7.21)	64.82 (8.03)

*The accuracy with * denotes that the method significantly outperforms other methods using paired t-test (p<0.05).*

The similar results are also shown in the MAHNOB-HCI database, as shown in Table X. The proposed HSLT and HSLT-Large are both significant outperformed other configurations in the arousal and valence classification. And the HSLT significantly outperforms HSLT-Large in the arousal classification. And the accuracy of the proposed HSLT improve 0.34% than the HSLT-Large in the valence classification. Apparently, the proposed HSLT has advantages over other configurations in the valence and arousal classification.

D. The Analysis of the Positional Embedding and Class Token

To investigate the effect of positional embedding (PE) and class token (denoted as [CLS]), we employ the ablation experiments and results are reported in the Table XI and Table XII. It can be seen that the performance of the proposed HSLT has significantly decreased (more than 5%) when the PE or [CLS] is removed from HSLT. What's more, the lowest accuracies have been obtained when the PE and [CLS] are both removed (decreased more than 6%). In summary, these results indicate that the PE and [CLS] are both contributed to improve the classification performance.

VI. DISCUSSION

The outstanding results achieved by proposed HSLT have shown the in the section V. In this part, the analysis of

表 VII
T R D N
MAHNOB-HCI D (%)

	LA vs. HA		LV vs. HV	
	P_{acc}	P_f	P_{acc}	P_f
The proposed HSLT	**66.20 (10.48)**	63.85 (10.23)	**66.63 (6.51)**	65.53 (7.21)
One-stage Transformer#	61.34 (10.85)	58.82 (10.02)	61.51 (7.17)	60.21 (7.20)
CNN* [34]	58.15 (10.62)	54.71 (10.54)	59.47 (5.96)	58.92 (6.62)
LSTM* [35]	56.84 (10.28)	53.87 (9.36)	57.98 (6.89)	53.67 (7.06)
DBN* [36]	56.95 (10.93)	53.31 (10.02)	59.78 (9.36)	58.59 (9.49)
CNN-LSTM* [37]	62.31 (9.07)	60.70 (8.53)	62.74 (8.21)	61.05 (8.69)
DenseNet* [38]	63.85 (9.15)	61.52 (8.76)	64.10 (8.62)	61.90 (7.99)

*The model with * denotes that we conduct same experiment by our own reimplementation.*

One-stage Transformer # means the all the 32 electrode patches are parallelly inputted into a transformer encoder. And the hyperparameters of one-stage HSLT are same to the transformer encoder of proposed HSLT in the brain-region-level spatial learning.

表 VIII
T H YPERPARAMETERS OF D HSLT C

	L_e	L_R	D_e	D_R	D_h	k
HSLT-Small	1	1	8	16	32	8
HSLT-Base	1	2	8	16	32	12
The proposed HSLT	2	2	8	16	64	16
HSLT-Large	2	3	8	16	64	16

C. 具有不同配置的 HSLT

在本部分，我们比较了不同配置的 HSLT 的性能。因此，我们提出了四种候选的 HSLT 配置，参数总结在表 VIII 中。这些配置表示为“HSLT-small”、“HSLT-base”、“提出的 HSLT”和“HSLT-Large”，其中深度和头数量逐渐增加。

不同 HSLT 配置在 DEAP 数据库中的性能列于表 IX 中。提出的 HSLT 在唤醒度和效价分类中实现了最佳性能，而 HSLT-Large 的性能与提出的 HSLT 相似。此外，与 HSLT-small 和 HSLT-base 相比，提出的 HSLT 和 HSLT-Large 通过配对 t 检验显著提高了准确率（$p < 0.05$）。这表明随着深度和头数量的增加，HSLT 能够学习到更具判别性的特征。此外，在效价分类中，提出的 HSLT 的准确率显著优于 HSLT-Large。在唤醒度分类中，提出的 HSLT 的准确率比 HSLT-Large 提高了 0.13%。

表 IX
T R D HSLT C
在 DEAP D (%)

	LA vs. HA		LV vs. HV	
	P_{acc}	P_f	P_{acc}	P_f
HSLT-Small	63.07 (10.49)	62.98 (8.90)	64.03 (8.40)	63.69 (7.75)
HSLT-Base	64.29 (10.85)	63.88 (10.02)	64.47 (8.36)	63.98 (7.42)
The proposed HSLT	**65.75 (8.51)**	64.29 (10.06)	**66.51* (8.53)**	66.27 (7.29)
HSLT-Large	65.62 (9.29)	64.18 (9.72)	65.50 (8.85)	64.62 (8.91)

表 X
T R D HSLT C ONFIGURATIONS IN
THE MAHNOB-HCI D (%)

	LA vs. HA		LV vs. HV	
	P_{acc}	P_f	P_{acc}	P_f
HSLT-Small	62.79 (10.72)	61.69 (10.37)	64.65 (6.69)	63.88 (8.60)
HSLT-Base	64.14 (11.79)	63.34 (12.32)	65.56 (6.74)	64.75 (8.23)
The proposed HSLT	**66.20* (10.48)**	63.85 (10.23)	**66.63 (6.51)**	65.53 (7.21)
HSLT-Large	64.01 (11.61)	62.91 (11.06)	66.29 (7.21)	64.82 (8.03)

*The accuracy with * denotes that the method significantly outperforms other methods using paired t-test (p<0.05).*

在 MAHNOB-HCI 数据库中也显示了相似的结果，如表 X 所示。所提出的 HSLT 和 HSLT-Large 在唤醒度和效价分类中均显著优于其他配置。并且 HSLT 在唤醒度分类中显著优于 HSLT-Large。在效价分类中，所提出的 HSLT 的准确率比 HSLT-Large 提高了 0.34%。显然，所提出的 HSLT 在效价和唤醒度分类中优于其他配置。

D. 位置嵌入和类标记的分析

为了研究位置嵌入（PE）和类标记（记为[CLS]）的影响，我们采用了消融实验，结果报告在表 XI 和表 XII 中。可以看出，当从 HSLT 中移除 PE 或[CLS]时，所提出的 HSLT 的性能显著下降（超过 5%）。此外，当同时移除 PE 和[CLS]时，获得了最低的准确率（下降超过 6%）。总之，这些结果表明 PE 和[CLS]都对提高分类性能做出了贡献。

VI. D

所提出的 HSLT 取得的突出结果已在第五部分展示。在这一部分，对

TABLE XI
THE RESULTS OF ABLATION EXPERIMENTS IN THE DEAP DATABASE (%)

	LA vs. HA		LV vs. HV	
	P_{acc}	P_f	P_{acc}	P_f
The proposed HSLT	**65.75*** **(8.51)**	64.29 (10.06)	**66.51*** **(8.53)**	66.27 (7.29)
Non-PE	62.45 (10.74)	61.22 (10.12)	63.15 (9.02)	61.84 (8.83)
Non-[CLS]	61.76 (10.99)	59.96 (10.86)	62.14 (8.51)	61.21 (7.64)
Non-PE and [CLS]	61.22 (9.60)	59.79 (9.66)	61.69 (8.41)	60.78 (7.86)

TABLE XII
THE RESULTS OF ABLATION EXPERIMENTS IN THE MAHNOB-HCI DATABASE (%)

	LA vs. HA		LV vs. HV	
	P_{acc}	P_f	P_{acc}	P_f
The proposed HSLT	**66.20*** **(10.48)**	63.85 (10.23)	**66.63*** **(6.51)**	65.53 (7.21)
Non-PE	59.67 (10.33)	58.89 (10.07)	60.41 (6.47)	59.45 (6.91)
Non-[CLS]	59.24 (9.74)	57.49 (9.34)	60.49 (7.07)	59.38 (7.47)
Non-PE and [CLS]	58.34 (8.86)	58.03 (8.96)	59.61 (5.88)	57.99 (5.64)

TABLE XIII
THE PERFORMANCE EVALUATION ON DIFFERENT FREQUENCY BANDS (%)

Frequency bands	DEAP LA vs. HA	DEAP LV vs. HV	MAHNOB-HCI LA vs. HA	MAHNOB-HCI LV vs. HV
Theta	58.69	58.67	59.03	58.45
Alpha	59.64	60.19	59.39	61.07
Slow alpha	59.73	61.32	59.51	61.15
Beta	59.98	61.98	60.72	61.77
Gamma	60.19	62.03	60.97	62.54
All bands	**65.75***	**66.51***	**66.20***	**66.63***

the frequency bands, analysis of brain regions, comparison with related researches are conducted to further verify the effectiveness of proposed HSLT.

A. The Analysis of the Frequency Bands

In this part, we conduct the performance evaluation to investigate relationship among frequency bands, and the results are shown in Table XIII. According to the results, the performance on Beta bands and Gamma bands is higher than the other bands in both arousal and valence classification. It indicates that the features from these two bands are more contributive to discriminate the emotional state. Moreover, the HSLT obtain optimal performance when using all bands features ($p < 0.05$). It suggests that the discriminative information is

TABLE XIV
THE PERFORMANCE EVALUATION ON DIFFERENT BRAIN REGIONS (%)

Brain regions	DEAP LA vs. HA	DEAP LV vs. HV	MAHNOB-HCI LA vs. HA	MAHNOB-HCI LV vs. HV
PF	61.25	61.58	61.96	62.13
F	61.04	61.71	60.91	62.41
LT	60.08	61.53	59.06	61.06
C	59.36	60.03	60.11	59.86
RT	60.29	61.02	59.34	61.92
LP	60.60	59.20	60.73	58.88
P	61.07	59.84	61.32	58.40
RP	60.76	60.61	62.07	58.06
O	59.75	60.90	58.38	58.26
All regions	**65.75***	**66.51***	**66.20***	**66.63***

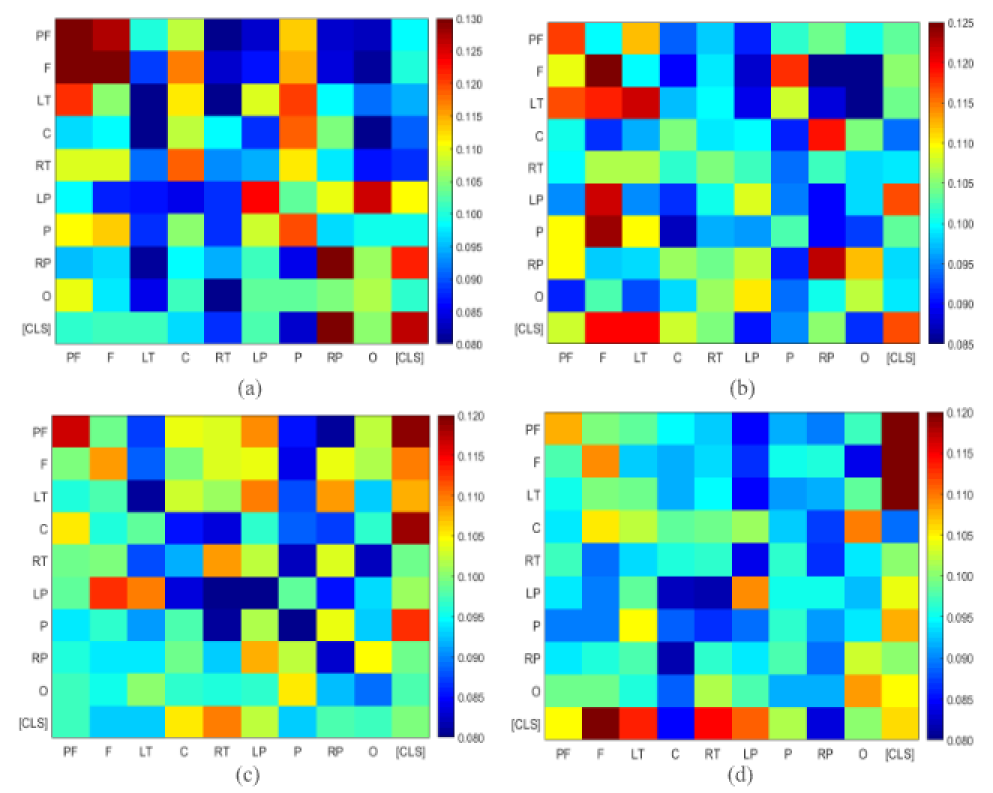

Fig. 3. The average score of the heads in the brain-region-level (a) subject25's arousal classification in the DEAP database (b) subject17's valence classification in the DEAP database (c) subject11's arousal classification in the MAHNOB-HCI database (d) subject6's valence classification in the MAHNOB-HCI database.

distributed in all the bands, and feature from different bands are complementary.

B. The Analysis of the Brain Regions

To analysis the essential brain regions, we firstly conduct the performance evaluation on each brain region, the results are shown in Table XIV. These results are obtained by the one-stage Transformer, because the HSLT is designed for capturing the dependencies among the multiple brain regions. According to the results in the Table XIV, the PF, F, and P achieve better performance on arousal classification in the DEAP database. Meanwhile, the PF, P and RP also achieves better performance in the MAHNOB-HCI database. It indicates that the pre-frontal lobe and parietal lobe are more contributive in the arousal classification. On the other hand, the PF, F, LT and RT achieve better performance on valence classification in both DEAP and MAHNOB-HCI database. It suggests that the pre-frontal lobe, frontal lobe and temporal lobe are more contributive in the valence classification.

表 XI

T R A E

在 DEAP D (%)

	LA vs. HA		LV vs. HV	
	P_{acc}	P_f	P_{acc}	P_f
The proposed HSLT	**65.75*** **(8.51)**	64.29 (10.06)	**66.51*** **(8.53)**	66.27 (7.29)
Non-PE	62.45 (10.74)	61.22 (10.12)	63.15 (9.02)	61.84 (8.83)
Non-[CLS]	61.76 (10.99)	59.96 (10.86)	62.14 (8.51)	61.21 (7.64)
Non-PE and [CLS]	61.22 (9.60)	59.79 (9.66)	61.69 (8.41)	60.78 (7.86)

表 XII

T R A E

THE MAHNOB-HCI D (%)

	LA vs. HA		LV vs. HV	
	P_{acc}	P_f	P_{acc}	P_f
The proposed HSLT	**66.20*** **(10.48)**	63.85 (10.23)	**66.63*** **(6.51)**	65.53 (7.21)
Non-PE	59.67 (10.33)	58.89 (10.07)	60.41 (6.47)	59.45 (6.91)
Non-[CLS]	59.24 (9.74)	57.49 (9.34)	60.49 (7.07)	59.38 (7.47)
Non-PE and [CLS]	58.34 (8.86)	58.03 (8.96)	59.61 (5.88)	57.99 (5.64)

TABLE XIII

T P E D

F B (%)

Frequency bands	DEAP		MAHNOB-HCI	
	LA vs. HA	LV vs. HV	LA vs. HA	LV vs. HV
Theta	58.69	58.67	59.03	58.45
Alpha	59.64	60.19	59.39	61.07
Slow alpha	59.73	61.32	59.51	61.15
Beta	59.98	61.98	60.72	61.77
Gamma	60.19	62.03	60.97	62.54
All bands	**65.75***	**66.51***	**66.20***	**66.63***

对频段、脑区分析以及与相关研究的比较，进一步验证了所提出的 HSLT 模型的有效性。

A. 频段分析

在本部分，我们进行性能评估以研究频段之间的关系，结果如表 XIII 所示。根据结果，在唤醒度和效价分类中，Beta 频段和 Gamma 频段的性能均高于其他频段。这表明这两个频段的特征更有助于区分情绪状态。此外，当使用所有频段特征时，HSLT 获得最佳性能（$p < 0.05$）。这表明判别信息是分布在所有频段中，不同频段的特征是互补的。

表 XIV

T P E D B R (%)

Brain regions	DEAP		MAHNOB-HCI	
	LA vs. HA	LV vs. HV	LA vs. HA	LV vs. HV
PF	61.25	61.58	61.96	62.13
F	61.04	61.71	60.91	62.41
LT	60.08	61.53	59.06	61.06
C	59.36	60.03	60.11	59.86
RT	60.29	61.02	59.34	61.92
LP	60.60	59.20	60.73	58.88
P	61.07	59.84	61.32	58.40
RP	60.76	60.61	62.07	58.06
O	59.75	60.90	58.38	58.26
All regions	**65.75***	**66.51***	**66.20***	**66.63***

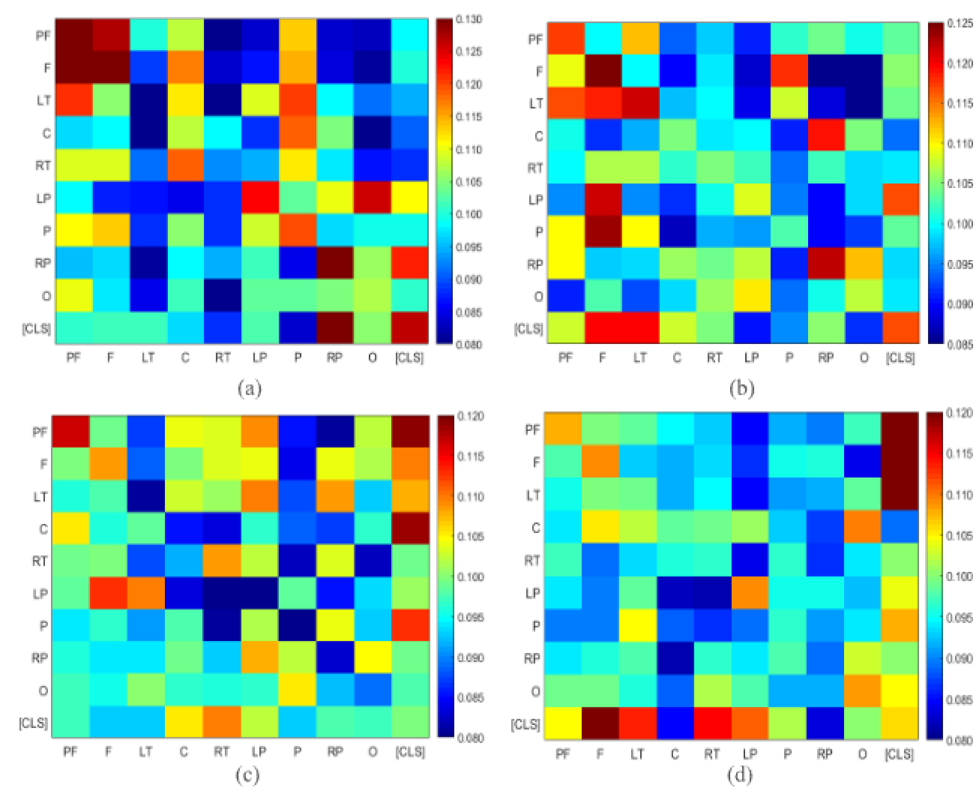

图 3. 大脑区域级别的头部平均得分（a）DEAP 数据库中受试者 25 的唤醒分类（b）DEAP 数据库中受试者 17 的效价分类（c）MAHNOB-HCI 数据库中受试者 11 的唤醒分类（d）MAHNOB-HCI 数据库中受试者 6 的效价分类。

B. 脑区分析

为分析关键脑区，我们首先对每个脑区进行性能评估，结果如表 XIV 所示。这些结果是通过单阶段 Transformer 获得的，因为 HSLT 的设计目的是捕捉多个脑区之间的依赖关系。根据表 XIV 中的结果，在 DEAP 数据库中，PF、F 和 P 在唤醒分类上表现更好。同时，在 MAHNOB-HCI 数据库中，PF、P 和 RP 也取得了更好的性能。这表明前额叶和顶叶在唤醒分类中贡献更大。另一方面，在 DEAP 和 MAHNOB-HCI 数据库中，PF、F、LT 和 RT 在效价分类上表现更好。这表明前额叶、额叶和颞叶在效价分类中贡献更大。

TABLE XV
THE DETAILS OF THE RELATED RESEARCHES USING SUBJECT-INDEPENDENT SCHEME

Related researches	Database	Number of classes & classifier	Validation method	P_{acc} (%)
Li *et al.* [40] (2018)	DEAP	2-classes, SVM	Leave-one-subject-out	Arousal: - Valence:59.06
Pandey *et al.* [41] (2019)	DEAP	2-classes, Deep neural network	30 subjects for training, 2 subjects for testing	Arousal:61.25 Valence:62.50
Rayatdoost *et al.* [39] (2018)	DEAP/ MAHNOB-HCI	2-classes Random Forest	Leave-one-subject-out	Arousal:59.22 Valence:55.70 (DEAP) Arousal:71.25 Valence:61.46 (MAHNOB-HCI)
Yin *et al.* [42] (2020)	DEAP/ MAHNOB-HCI	2-classes, LSSVM	Leave-one-subject-out	Arousal:65.10 Valence:67.97 (DEAP) Arousal:67.43 Valence:70.90 (MAHNOB-HCI)
Zhang *et al.* [43] (2020)	DEAP/ MAHNOB-HCI	2-classes, KNN	Leave-one-subject-out	Arousal:65.21 Valence:66.35 (DEAP) Arousal:65.20 Valence:65.37 (MAHNOB-HCI)
Huang *et al.* [44] (2016)	MAHNOB-HCI	3-classes, SVM	Leave-one-subject-out	Arousal: 62.00 Valence: 61.00
Soleymani *et al.* [27] (2012)	MAHNOB-HCI	3-classes, SVM	Leave-one-subject-out	Arousal: 52.40 Valence: 57.00
The proposed HSLT	DEAP/ MAHNOB-HCI	2-classes, HSLT	Leave-one-subject-out	Arousal: 65.75 Valence: 66.51 (DEAP) Arousal:66.20 Valence:66.63 (MAHNOB-HCI)

To further analysis the brain regions in the HSLT, we visualize the heads within the self-attention. The average scores of the heads are illustrated in the Fig. 3. Each column can be seen as the contribution of the corresponding brain region or [CLS] in the classification. In the DEAP database, we find that the pre-frontal lobe and frontal lobe (PF and F in the Fig. 3(a) and Fig. 3(b)) have achieved more contributions than other regions. And similar finding could be observed in the valence recognition in the MAHNOB-HCI database (PF and F in the Fig. 3(c)). Our findings basically coincide with the observation of in the neuroscience, pre-frontal lobe and frontal lobe are closely correlated to the emotions [12]. And the parietal lobe (P in the Fig. 3(a), LP and RP in the Fig. 3(c)) achieves more contribution in the arousal recognition. This is basically consistent with the finding in the neuroscience study [13], the parietal lobe is associated with arousal changes in humans. Similarly, the findings of temporal lobe (LT in the Fig. 3(b) and Fig. 3(d)) in the valence recognition are also basically consistent with the findings of Koelstra *et al.* [26]. Besides, we notice a significant contribution from class token in the MAHNOB-HCI database ([CLS] in the Fig. 3(b) and Fig. 3(d)). It could verify the importance of the class token. Overall, these results indicate that the self-attention within HSLT could learn the discriminate features from the critical brain regions.

Furthermore, the Fig. 3. have shown the advantages of HSLT, and it also shown the difficulty in the EEG-based emotion recognition. We find the inconsistent emotional pattern between different subjects and databases. It is might associated with sensory information which is generated by different stimulus materials (Music videos in DEAP and movie clips in MAHNOB-HCI) and individual-specific differences [39].

C. Comparison With Related Researches

We compare the performance of HSLT with recently researches using subject-independent scheme in the DEAP and MAHNOB-HCI database. The performances of different methods are listed in the Table XV. Except for the Ref. [41], all the studies adopt leave-one-subject-out cross-validation paradigm. The accuracy of HSLT is higher than the results in [27], [40], [41], [43] and [44]. And our method outperforms the Ref. [39] in the DEAP database and valence recognition in the MAHNOB-HCI. Compared with Ref. [42], our method is higher than it in the arousal recognition of DEAP database, but the performances of our method are less in the other tasks. Overall, the proposed HSLT has achieved the outstanding performance among these methods.

Compared with these works, the advantages of HSLT can be concluded as follows. Firstly, the self-attention mechanism emphasizes the contributive brain regions and enhances spatial

表 XV

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Related researches	Database	Number of classes & classifier	Validation method	P_{acc} (%)
Li *et al.* [40] (2018)	DEAP	2-classes, SVM	Leave-one-subject-out	Arousal: - Valence:59.06
Pandey *et al.* [41] (2019)	DEAP	2-classes, Deep neural network	30 subjects for training, 2 subjects for testing	Arousal:61.25 Valence:62.50
Rayatdoost *et al.* [39] (2018)	DEAP/ MAHNOB-HCI	2-classes Random Forest	Leave-one-subject-out	Arousal:59.22 Valence:55.70 (DEAP) Arousal:71.25 Valence:61.46 (MAHNOB-HCI)
Yin *et al.* [42] (2020)	DEAP/ MAHNOB-HCI	2-classes, LSSVM	Leave-one-subject-out	Arousal:65.10 Valence:67.97 (DEAP) Arousal:67.43 Valence:70.90 (MAHNOB-HCI)
Zhang *et al.* [43] (2020)	DEAP/ MAHNOB-HCI	2-classes, KNN	Leave-one-subject-out	Arousal:65.21 Valence:66.35 (DEAP) Arousal:65.20 Valence:65.37 (MAHNOB-HCI)
Huang *et al.* [44] (2016)	MAHNOB-HCI	3-classes, SVM	Leave-one-subject-out	Arousal: 62.00 Valence: 61.00
Soleymani *et al.* [27] (2012)	MAHNOB-HCI	3-classes, SVM	Leave-one-subject-out	Arousal: 52.40 Valence: 57.00
The proposed HSLT	DEAP/ MAHNOB-HCI	2-classes, HSLT	Leave-one-subject-out	Arousal: 65.75 Valence: 66.51 (DEAP) Arousal:66.20 Valence:66.63 (MAHNOB-HCI)

为了进一步分析 HSLT 中的脑区，我们可视化了自注意力机制中的头。每个头的平均分数如图 3 所示。每一列可以看作是相应脑区或 [CLS] 在分类中的贡献。在 DEAP 数据库中，我们发现前额叶皮层和额叶（图 3(a) 和图 3(b) 中的 PF 和 F）的贡献度高于其他区域。在 MAHNOB-HCI 数据库的效价识别中也可以观察到类似发现（图 3(c) 中的 PF 和 F）。我们的发现基本上与神经科学中的观察结果一致，前额叶皮层和额叶与情绪密切相关 [12]。而顶叶（图 3(a) 中的 P，图 3(c) 中的 LP 和 RP）在唤醒识别中的贡献度更高。这与神经科学研究中的发现基本一致 [13]，顶叶与人类唤醒变化相关。同样，颞叶（图 3(b) 和图 3(d) 中的 LT）在效价识别中的发现也与 Koelstra 等人的研究 [26] 基本一致。 此外，我们注意到在 MAHNOB-HCI 数据库中类标记（[CLS]在图 3(b)和图 3(d)中）有显著贡献。这可以验证类标记的重要性。总体而言，这些结果表明 HSLT 中的自注意力机制能够从关键脑区学习到区分性特征。

此外，图 3 展示了 HSLT 的优势，同时也显示了基于脑电的情绪识别的难度。我们发现不同受试者和数据库之间存在不一致的情绪模式。这可能与由不同刺激材料（DEAP 中的音乐视频和 MAHNOB-HCI 中的电影片段）产生的感官信息以及个体特异性差异[39]有关。

C. 与相关研究的比较

我们比较了 HSLT 与最近使用受试者独立方案在 DEAP 和 MAHNOB-HCI 数据库中的研究性能。不同方法的性能列于表 XV 中。除了参考文献[41]，所有研究都采用留一受试者交叉验证范式。HSLT 的准确率高于[27]、[40]、[41]、[43]和[44]中的结果。在我们的方法在 DEAP 数据库的效价识别和 MAHNOB-HCI 的效价识别中优于参考文献[39]。与参考文献[42]相比，我们的方法在 DEAP 数据库的唤醒识别中表现更高，但在其他任务中的性能较低。总体而言，所提出的 HSLT 在这些方法中取得了出色的性能。

与这些工作相比，HSLT 的优势可以总结如下。首先，自注意力机制强调了贡献性脑区并增强了空间

dependencies capturing. Next, the hierarchical learning strategy could effectively integrate the spatial information within the brain regions and capture the dependencies among the brain regions. Finally, the PE and [CLS] retain the positional information and representative information which are essential for the sequence learning.

Meanwhile, the limitation of HSLT is also valuable to discuss. It is difficult to acquire a large amount of emotional EEG data [45]. However, transformer encoder needs the sufficient data for adequately revealing the strong ability of sequence learning. Although we have adopted sliding window to divide a trail into several segments for obtaining more samples, the data size restricts the performance of HSLT to some extent. For example, the performance would be slightly decrease when we design a deeper network than the proposed HSLT (in the Section V *HSLT with different configurations*).

VII. Conclusion

In this work, we propose a novel HSLT model to robustly learn the EEG spatial information and apply it to achieve arousal and valence recognition. The PSD features are adopted to evaluate the HSLT performance, and the HSLT has achieved outstanding performance in subject-independent experiment with the accuracy of arousal and valence 65.75%/66.20% and 66.51%/66.63% in the DEAP and MAHNOB-HCI database respectively. We conduct extensive experiments to verify the effectiveness of HSLT. What's more, the visualization of the self-attention demonstrates that the self-attention within HSLT could emphasize discriminative information from contributive brain regions. We also find that positional embedding and class token also make contributions to boosting the performance. In addition, we also analysis the limitation of HSLT. In the future work, the data augmentation algorithms for EEG signals are necessary to investigate for further improving the emotion recognition performance.

References

[1] P. Leelaarporn *et al.*, "Sensor-driven achieving of smart living: A review," *IEEE Sensors J.*, vol. 21, no. 9, pp. 10369–10391, May 2021.

[2] R. W. Picard, *Affective Computing*, vol. 604. Cambridge, MA, USA: MIT Press, 1997, pp. 247–248.

[3] F. Demir, N. Sobahi, S. Siuly, and A. Sengur, "Exploring deep learning features for automatic classification of human emotion using EEG rhythms," *IEEE Sensors J.*, vol. 21, no. 13, pp. 14923–14930, Jul. 2021.

[4] S. K. Khare and V. Bajaj, "An evolutionary optimized variational mode decomposition for emotion recognition," *IEEE Sensors J.*, vol. 21, no. 2, pp. 2035–2042, Jan. 2021.

[5] P. Lakhan *et al.*, "Consumer grade brain sensing for emotion recognition," *IEEE Sensors J.*, vol. 19, no. 21, pp. 9896–9907, Nov. 2019.

[6] L. Kulke, D. Feyerabend, and A. Schacht, "A comparison of the affectiva iMotions facial expression analysis software with EMG for identifying facial expressions of emotion," *Frontiers Psychol.*, vol. 11, p. 329, Feb. 2020.

[7] F. Agrafioti, D. Hatzinakos, and A. K. Anderson, "ECG pattern analysis for emotion detection," *IEEE Trans. Affect. Comput.*, vol. 3, no. 1, pp. 102–115, Jan. 2012.

[8] H. Gunes and B. Schuller, "Categorical and dimensional affect analysis in continuous input: Current trends and future directions," *Image Vis. Comput.*, vol. 31, no. 2, pp. 120–136, Feb. 2013.

[9] X. W. Wang, D. Nie, and B. L. Lu, "Emotional state classification from EEG data using machine learning approach," *Neurocomputing*, vol. 129, no. 10, pp. 94–106, Apr. 2014.

[10] E. Kroupi, J. M. Vesin, and T. Ebrahimi, "Subject-independent odor pleasantness classification using brain and peripheral signals," *IEEE Trans. Affect. Comput.*, vol. 7, no. 4, pp. 422–434, Oct./Dec. 2016.

[11] S. K. Khare and V. Bajaj, "Time–frequency representation and convolutional neural network-based emotion recognition," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 7, pp. 2901–2909, Jul. 2021.

[12] A. Etkin, T. Egner, and R. Kalisch, "Emotional processing in anterior cingulate and medial prefrontal cortex," *Trends Cogn. Sci.*, vol. 15, no. 2, pp. 85–93, Feb. 2011.

[13] S. Anders, M. Lotze, M. Erb, W. Grodd, and N. Birbaumer, "Brain activity underlying emotional valence and arousal: A response-related fMRI study," *Hum. Brain Mapping*, vol. 23, no. 4, pp. 200–209, 2004.

[14] S. Sangnark *et al.*, "Revealing preference in popular music through familiarity and brain response," *IEEE Sensors J.*, vol. 21, no. 13, pp. 14931–14940, Jul. 2021.

[15] A. Dosovitskiy *et al.*, "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. 9th Int. Conf. Learn. Represent. (ICLR)*, Jan. 2021, pp. 1–21.

[16] R. Khosrowabadi, C. Quek, K. K. Ang, and A. Wahab, "ERNN: A biologically inspired feedforward neural network to discriminate emotion from EEG signal," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 25, no. 3, pp. 609–620, Mar. 2014.

[17] P. Bashivan, I. Rish, M. Yeasin, and N. Codella, "Learning representations from EEG with deep recurrent-convolutional neural networks," in *Proc. 4th Int. Conf. Learn. Represent. (ICLR)*, May 2016, pp. 1–15.

[18] T. Zhang, W. Zheng, Z. Cui, Y. Zong, and Y. Li, "Spatial–temporal recurrent neural network for emotion recognition," *IEEE Trans. Cybern.*, vol. 49, no. 3, pp. 839–847, Mar. 2019.

[19] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 7794–7803.

[20] W. Tao *et al.*, "EEG-based emotion recognition via channel-wise attention and self attention," *IEEE Trans. Affect. Comput.*, early access, Sep. 22, 2020, doi: 10.1109/TAFFC.2020.3025777.

[21] J. X. Chen, D. M. Jiang, and Y. N. Zhang, "A hierarchical bidirectional GRU model with attention for eeg-based emotion classification," *IEEE Access*, vol. 7, pp. 118530–118540, 2019.

[22] A. Vaswani *et al.*, "Attention is all you need," in *Proc. Ann. Conf. Neural Inf. Process. Syst. (NIPS)*, Dec. 2017, pp. 5998–6008.

[23] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics, Hum. Lang. Technol. (HAACL-HLT)*, Jun. 2019, pp. 4171–4186.

[24] J. Sun, J. Xie, and H. Zhou, "EEG classification with transformer-based models," in *Proc. IEEE 3rd Global Conf. Life Sci. Technol. (LifeTech)*, Mar. 2021, pp. 92–93.

[25] J. Pedoeem, S. Abittan, G. B. Yosef, and S. Keene, "TABS: Transformer based seizure detection," in *Proc. IEEE Signal Process. Med. Biol. Symp. (SPMB)*, Dec. 2020, pp. 1–6.

[26] S. Koelstra *et al.*, "DEAP: A database for emotion analysis using physiological signals," *IEEE Trans. Affect. Comput.*, vol. 3, no. 1, pp. 18–31, Jan. 2012.

[27] M. Soleymani, J. Lichtenauer, T. Pun, and M. Pantic, "A multimodal database for affect recognition and implicit tagging," *IEEE Trans. Affect. Comput.*, vol. 3, no. 1, pp. 42–55, Jan. 2012.

[28] A. Delorme and S. Makeig, "EEGLAB: An open source toolbox for analysis of single-trial EEG dynamics including independent component analysis," *J. Neurosci. Methods*, vol. 134, no. 1, pp. 9–21, Mar. 2004.

[29] K. A. Lindquist, T. D. Wager, H. Kober, E. Bliss-Moreau, and L. F. Barret, "The brain basis of emotion: A meta-analytic review," *Behav. Brain Sci.*, vol. 35, no. 3, pp. 121–143, Jun. 2012.

[30] K. Vytal and S. Hamann, "Neuroimaging support for discrete neural correlates of basic emotions: A voxel-based meta-analysis," *J. Cogn. Neurosci.*, vol. 22, no. 12, pp. 2864–2885, Dec. 2010.

[31] G. C. Ribas, "The cerebral sulci and gyri," *Neurosurg. Focus*, vol. 28, no. 2, p. E2, Feb. 2010.

[32] W. Song *et al.*, "AutoInt: Automatic feature interaction learning via self-attentive neural networks," in *Proc. 28th ACM Int. Conf. Inf. Knowl. Manage. (CIKM)*, Nov. 2019, pp. 1161–1170.

[33] J. Lei Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," 2016, *arXiv:1607.06450*.

[34] P. Bashivan, I. Rish, M. Yeasin, and N. Codella, "Learning representations from EEG with deep recurrent-convolutional neural networks," in *Proc. 4th Int. Conf. Learn. Represent. (ICLR)*, 2016, pp. 1–15.

[35] S. Alhagry, A. A. Fahmy, and R. A. El-Khoribi, "Emotion recognition based on EEG using LSTM recurrent neural network," *Int. J. Adv. Comput. Sci. Appl.*, vol. 8, no. 10, pp. 355–358, 2017.

依赖关系捕获。接下来，分层学习策略能够有效地整合大脑区域内的空间信息，并捕获大脑区域之间的依赖关系。最后，PE 和[CLS]保留了位置信息和代表性信息，这些信息对于序列学习至关重要。

与此同时，讨论 HSLT 的局限性也是有益的。获取大量情绪脑电数据[45]是困难的。然而，Transformer 编码器需要足够的数据来充分揭示序列学习的强大能力。尽管我们采用了滑动窗口将一个轨迹分成几个片段以获得更多样本，但数据量在一定程度上限制了 HSLT 的性能。例如，当设计比所提出的 HSLT 更深层的网络时（在第五节不同配置的 HSLT），性能会略有下降。

VII. C

在这项工作中，我们提出了一种新的 HSLT 模型，用于稳健地学习脑电图空间信息，并将其应用于实现唤醒度和效价识别。采用 PSD 特征来评估 HSLT 的性能，HSLT 在受试者无关的实验中取得了优异的性能，在 DEAP 和 MAHNOB-HCI 数据库中唤醒度和效价的准确率分别为 65.75%/66.20%和 66.51%/66.63%。我们进行了广泛的实验来验证 HSLT 的有效性。此外，自注意力的可视化表明，HSLT 中的自注意力能够强调来自贡献性脑区的判别性信息。我们还发现位置嵌入和类别标记也对提升性能做出了贡献。此外，我们还分析了 HSLT 的局限性。在未来工作中，有必要研究脑电图信号的数据增强算法，以进一步提高情绪识别性能。

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[1] P. Leelaarporn 等人，“基于传感器的智能生活实现：综述，”IEEE 传感器杂志，第 21 卷，第 9 期，第 10369–10391 页，2021 年 5 月。[2] R. W. Picard，《情感计算》，第 604 卷。美国马萨诸塞州剑桥：麻省理工学院出版社，1997 年，第 247–248 页。

[3] F. Demir, N. Sobahi, S. Siuly 和 A. Sengur，“探索深度学习特征用于脑电波节律的人类情感自动分类，”IEEE 传感器杂志，第 21 卷，第 13 期，第 14923–14930 页，2021 年 7 月。[4] S. K. Khare 和 V. Bajaj，“一种进化优化的变分模态分解用于情感识别，”IEEE 传感器杂志，第 21 卷，第 2 期，第 2035–2042 页，2021 年 1 月。

[5] P. Lakhan 等人，“消费级脑部传感用于情绪识别，”IEEE 传感器杂志，第 19 卷，第 21 期，第 9896–9907 页，2019 年 11 月。[6] L. Kulke, D. Feyerabend 和 A. Schacht，“情感分析软件 iMotions 面部表情分析与 EMG 在识别情绪面部表情方面的比较，”心理学前沿，第 11 卷，第 329 页，2020 年 2 月。[7] F. Agrafioti, D. Hatzinakos 和 A. K. Anderson，“心电图模式分析用于情绪检测，”IEEE 情感计算汇刊，第 3 卷，第 1 期，第 102–115 页，2012 年 1 月。[8] H. Gunes 和 B. Schuller，“连续输入中的类别和维度情感分析：当前趋势和未来方向，”图像视觉。

Comput., vol. 31, no. 2, pp. 120–136, Feb. 2013.

[9] 王晓伟, 聂栋, 和 陆宝莲, “基于机器学习的脑电数据情绪状态分类,” 神经计算, 卷. 129, 期. 10, 页. 94–106, 2014 年 4 月.

[10] E. Kroupi, J. M. Vesin, and T. Ebrahimi, “Subject-independent odor 基于脑部和外周信号的情绪愉悦度分类，”IEEE 情感计算汇刊，第 7 卷，第 4 期，第 422-434 页，2016 年 10 月/12 月。

[11] S. K. Khare 和 V. Bajaj，“时频表示和卷积 基于深度神经网络的情感识别，”IEEE Transactions on Neural Networks and Learning Systems，第 32 卷，第 7 期，第 2901-2909 页，2021 年 7 月

[12] A. Etkin, T. Egner, and R. Kalisch, “情绪处理在前额叶 cingulate 和内侧前额叶皮层,” Trends Cogn. Sci., vol. 15, no. 2, pp. 85–93, Feb. 2011.

[13] S. Anders, M. Lotze, M. Erb, W. Grodd, and N. Birbaumer, “Brain 活动在情绪效价和唤醒性下的基础：一项与反应相关的 fMRI 研究,” Hum. Brain Mapping, vol. 23, no. 4, pp. 200–209, 2004.

[14] S. Sangnark 等，“通过” 熟悉度和脑部反应，”IEEE 传感器杂志，第 21 卷，第 13 期，第 14931-14940 页，2021 年 7 月。

[15] A. Dosovitskiy 等人，“一张图值 16×16 个词：用于大规模图像识别的 Transformer”，在第九届国际学习表示会议论文集 (ICLR), 2021 年 1 月, 第 1-21 页。

[16] R. Khosrowabadi, C. Quek, K. K. Ang, 和 A. Wahab, “ERNN: 一种受生物学启发的前馈神经网络，用于从脑电图信号中识别情绪，”IEEE 神经网络与学习系统汇刊，第 25 卷，第 3 期，第 609-620 页，2014 年 3 月。

[17] P. Bashivan, I. Rish, M. Yeasin, 和 N. Codella, “学习表示 从脑电图（EEG）中提取情绪信息，使用深度循环-卷积神经网络，”在 2016 年第四届国际学习表征会议（ICLR）论文集中，5 月，第 1-15 页。

[18] 张涛，郑伟，崔峥，宗宇，李岩，“时空 用于情绪识别的循环神经网络，”IEEE 资讯科学汇刊，第 49 卷，第 3 期，第 839-847 页，2019 年 3 月。

[19] 王翔，R. Girshick，A. Gupta 和 K. He，“非局部神经网络，”在 IEEE/CVF 计算机视觉与模式识别会议论文集中 (CVPR), 2018 年 6 月, 第 7794–7803 页。

[20] W. Tao 等人，“基于脑电的情绪识别：通道级注意力” 注意和自注意力，”IEEE 情感计算汇刊，早期访问，2020 年 9 月 22 日，doi: 10.1109/TAFFC.2020.3025777.

[21] J. X. Chen, D. M. Jiang, 和 Y. N. Zhang, “一种用于基于脑电图的情绪分类的具有注意力的层次双向 GRU 模型,” IEEE Access, 第 7 卷, 第 118530–118540 页, 2019.

[22] A. Vaswani 等, “Attention is all you need,” 在 2017 年 12 月神经信息处理系统年度会议论文集 (NIPS) 中, 第 5998–6008 页.

[23] J. Devlin, M. Chang, K. Lee, 和 K. Toutanova, “BERT: 用于语言理解的深度双向 Transformer 的预训练,” 在北美计算机语言学协会分会会议论文集中, Technol. (NAACL-HLT), 2019 年 6 月, 第 4171–4186 页。

[24] 孙杰，谢俊，周航，“基于 Transformer 的脑电图分类 “模型,”在 IEEE 第三届全球生命科学技术大会（LifeTech）论文集中，2021 年 3 月，第 92-93 页。

[25] J. Pedoeem, S. Abittan, G. B. Yosef, 和 S. Keene, “TABS: 基于 Transformer 的癫痫检测,” 在 IEEE 信号处理医学生物学研讨会论文集中 (SPMB), 2020 年 12 月, 第 1-6 页。

[26] S. Koelstra 等人，“DEAP：一个用于情绪分析的数据库” 生理信号，”IEEE 情感计算汇刊，第 3 卷，第 1 期，第 18–31 页，2012 年 1 月。

[27] M. Soleymani, J. Lichtenauer, T. Pun, 和 M. Pantic, “一个用于情感识别和隐式标记的多模态数据库,” IEEE 情感计算汇刊 Comput., 卷 3, 第 1 期, 第 42–55 页, 2012 年 1 月。

[28] A. Delorme 和 S. Makeig，“EEGLAB: 一个开源工具- 用于单次试验脑电图动力学分析的独立成分分析框，” J. Neurosci. Methods, vol. 134, no. 1, pp. 9–21, Mar. 2004.

[29] K. A. Lindquist, T. D. Wager, H. Kober, E. Bliss-Moreau, and L. F. Barret, “情绪的脑基础：一项元分析综述,” Behav. Brain Sci., vol. 35, no. 3, pp. 121–143, Jun. 2012.

[30] K. Vytal 和 S. Hamann, “神经影像学支持基本情绪的离散神经相关物：基于体素的元分析,” J. Cogn. Neurosci., vol. 22, no. 12, pp. 2864–2885, Dec. 2010.

[31] G. C. Ribas, “大脑的脑沟和脑回,” Neurosurg. Focus, vol. 28, no. 2, p. E2, Feb. 2010. [32] W. Song 等人, “AutoInt: 基于自注意力神经网络的自动特征交互学习,” 在第 28 届 ACM 国际信息知识会议论文集中

管理。(CIKM), 2019 年 11 月, 第 1161-1170 页。

[33] J. Lei Ba, J. R. Kiros, 和 G. E. Hinton, “Layer normalization,” 2016, arXiv:1607.06450.

[34] P. Bashivan, I. Rish, M. Yeasin, 和 N. Codella, “学习表示 从脑电图（EEG）中提取情绪信息，使用深度循环-卷积神经网络，”在 2016 年第四届国际学习表征会议（ICLR）论文集中，第 1-15 页。

[35] S. Alhagry, A. A. Fahmy, 和 R. A. El-Khoribi, “基于脑电图的情绪识别，使用 LSTM 循环神经网络,” 国际先进期刊 计算机科学应用，第 8 卷，第 10 期，第 355-358 页，2017 年。

[36] W.-L. Zheng, J.-Y. Zhu, Y. Peng, and B.-L. Lu, "EEG-based emotion classification using deep belief networks," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2014, pp. 1–6.

[37] Z. Gao, T. Yuan, X. Zhou, C. Ma, K. Ma, and P. Hui, "A deep learning method for improving the classification accuracy of SSMVEP-based BCI," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 67, no. 12, pp. 3447–3451, Dec. 2020.

[38] Z. Gao, X. Wang, Y. Yang, Y. Li, K. Ma, and G. Chen, "A channel-fused dense convolutional network for EEG-based emotion recognition," *IEEE Trans. Cogn. Develop. Syst.*, vol. 13, no. 4, pp. 945–954, Dec. 2021.

[39] S. Rayatdoost and M. Soleymani, "Cross-corpus EEG-based emotion recognition," in *Proc. IEEE 28th Int. Workshop Mach. Learn. Signal Process. (MLSP)*, Sep. 2018, pp. 1–6.

[40] X. Li, D. Song, P. Zhang, Y. Zhang, Y. Hou, and B. Hu, "Exploring EEG features in cross-subject emotion recognition," *Frontiers Neurosci.*, vol. 12, p. 162, May 2018.

[41] P. Pandey and K. R. Seeja, "Subject independent emotion recognition from EEG using VMD and deep learning," *J. King Saud Univ. Comput. Inf. Sci.*, Nov. 2019.

[42] Z. Yin, L. Liu, J. Chen, B. Zhao, and Y. Wang, "Locally robust EEG feature selection for individual-independent emotion recognition," *Expert Syst. Appl.*, vol. 162, Dec. 2020, Art. no. 113768.

[43] W. Zhang, Z. Yin, Z. Sun, Y. Tian, and Y. Wang, "Selecting transferrable neurophysiological features for inter-individual emotion recognition via a shared-subspace feature elimination approach," *Comput. Biol. Med.*, vol. 123, Aug. 2020, Art. no. 103875.

[44] X. Huang *et al.*, "Multi-modal emotion analysis from facial expressions and electroencephalogram," *Comput. Vis. Image Understand.*, vol. 147, pp. 114–124, Jun. 2016.

[45] W. Li, W. Huan, B. Hou, Y. Tian, Z. Zhang, and A. Song, "Can emotion be transferred?—A review on transfer learning for EEG-based emotion recognition," *IEEE Trans. Cogn. Develop. Syst.*, early access, Jul. 21, 2021, doi: 10.1109/TCDS.2021.3098842.

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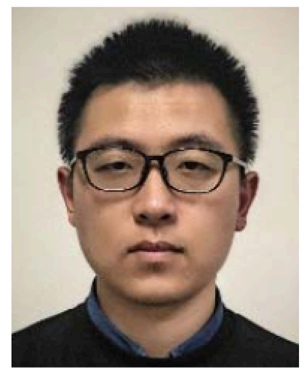

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[36] 郑伟立, 朱建勇, 彭勇, 和 陆冰立, “基于脑电的情绪分类使用深度信念网络,” 在 IEEE 国际会议论文集
多媒体博览会（ICME），2014 年 7 月，第 1-6 页。
[37] 高志， 袁涛， 周翔， 马超， 马凯， 和惠平， “一种深度
一种用于提高基于 SSMVEP 的 BCI 分类准确性的学习方法,”IEEE 电路与系统 II：实验简报，第 67 卷，第 12 期，第 3447-3451 页，2020 年 12 月。
[38] 高志，王翔，杨勇，李岩，马凯，和陈刚，“一种通道融合
基于脑电图的情绪识别的密集卷积网络,”IEEE 认知发展系统汇刊，第 13 卷，第 4 期，第 945-954 页，2021 年 12 月。
[39] S. Rayatdoost 和 M. Soleymani， “跨语料库的基于脑电图的情绪
"情绪识别,”在 IEEE 第 28 届国际机器学习信号处理研讨会（MLSP）论文集中，2018 年 9 月，第 1-6 页。
[40] 李翔, 宋东, 张鹏, 张宇, 侯宇, 和 胡博, “探索
基于脑电图的特征在跨被试情绪识别中,”神经科学前沿，第 12 卷，第 162 页，2018 年 5 月。
[41] P. Pandey 和 K. R. Seeja, “基于 EEG 的无受试者情绪识别：使用 VMD 和深度学习,” J. King Saud Univ. Comput.
《信息科学》, 2019 年 11 月。
[42] Z. Yin, L. Liu, J. Chen, B. Zhao, 和 Y. Wang, “针对个体独立情绪识别的鲁棒 EEG 特征选择方法,”
Expert Syst. Appl., vol. 162, Dec. 2020, Art. no. 113768.
[43] 张伟，尹哲，孙哲，田宇，王宇，“选择可迁移
基于共享子空间特征消除方法的人体间情绪识别的神经生理特征,” 计算生物学与医学，第 123 卷，2020 年 8 月，文章编号 103875。
[44] 黄翔等， “基于面部表情的多模态情感分析
和脑电图,” 计算机视觉与图像理解，vol. 147，pp. 114–124，2016 年 6 月。
[45] 李伟，黄伟，侯斌，田宇，张哲，宋安，“能否
情绪能否迁移？——基于脑电图的情绪识别迁移学习综述,”IEEE 认知发展系统汇刊，早期访问，2021 年 7 月 21 日，doi: 10.1109/TCDS.2021.3098842.

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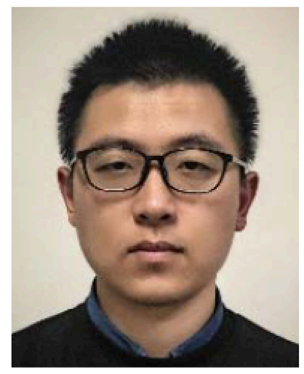

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